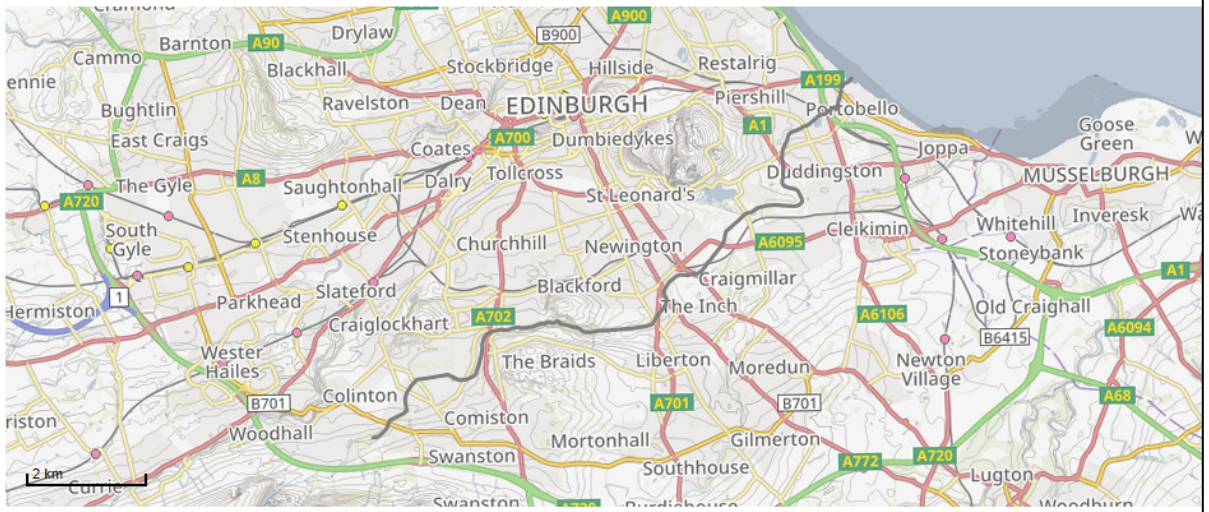
# SEPA Water Classification Hub – Braid Burn

Search information sheets: Search ▶ Return to summary tool

Tool Guidance
Terms and Conditions

Braid Burn (Upstream Dreghorn Barracks to Portobello) is a river (ID: 3500), in the Edinburgh Coastal catchment of the Scotland river basin district. The main stem is approximately 12.6 kilometres in length. The water body has been designated as a heavily modified water body on account of physical alterations that cannot be addressed without a significant impact from an increased risk of subsidence or flooding.




## Water classification data for selected water body

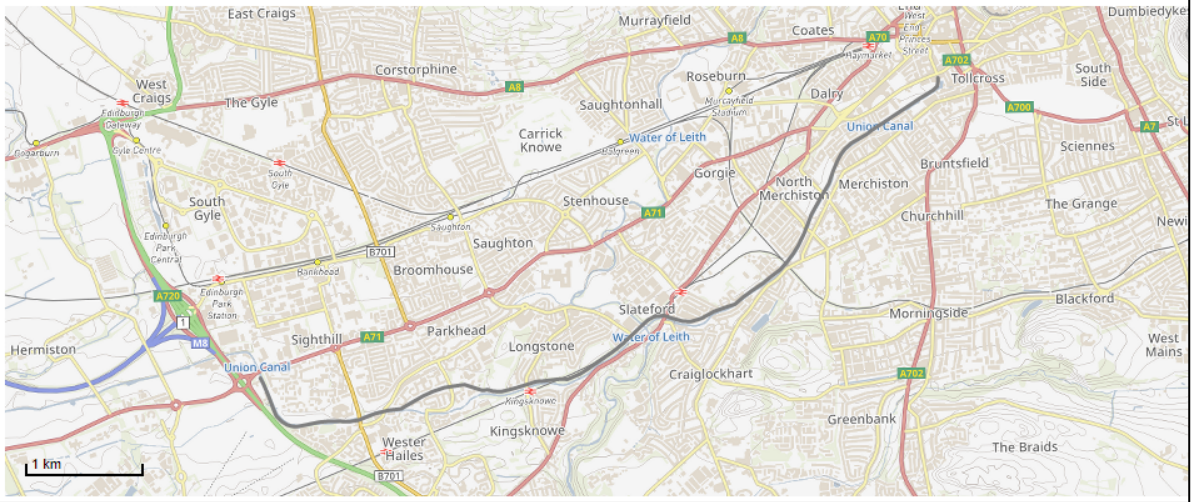
| ID | Name | Parameter | 2020 | 2019 | 2018 | 2017 | 2016 | 2015 | 2014 | 2013 | 2012 | 2011 | 2010 | 2009 | 2008 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 3500 | Braid Burn (Upstream Dreghorn Barracks to Portobello) | 1: Overall status | Moderate ecol… | Moderate ecol… | Bad ecological… | Bad ecological… | Bad ecological… | Bad ecological… | Bad ecological… | Bad ecological… | Bad ecological… | Bad ecological… | Bad ecological… | Bad ecological… | Bad ecological… | Poor |
| | | 1-1: Pre-HMWB status | Bad | Bad | Bad | Bad | Bad | Bad | Bad | Bad | Bad | Bad | Bad | Bad | Bad | |
| | | 1-3: Overall ecology | Bad | Bad | Bad | Bad | Bad | Bad | Bad | Bad | Bad | Bad | Bad | Bad | Bad | Poor |

# SEPA Water Classification Hub – Union Canal

Search information sheets: Search ▶ Return to summary tool

Tool Guidance
Terms and Conditions

Union canal (Murray Burn to Fountainbridge) is a river (ID: 9), in the Water of Leith catchment of the Scotland river basin district. The main stem is approximately 7.3 kilometres in length. The water body has been designated as an artifical water body on account of physical alterations that cannot be addressed without a significant impact on navigation.




## Water classification data for selected water body

| ID | Name | Parameter | 2020 | 2019 | 2018 | 2017 | 2016 | 2015 | 2014 | 2013 | 2012 | 2011 | 2010 | 2009 | 2008 | 20 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 9 | Union canal (Murray Burn to Fountainbridge) | 1: Overall status | Good ecologic… | Good ecologic… | Good ecologic… | Good ecologic… | Good ecologic… | Good ecologic… | Good ecologic… | Good ecologic… | Good ecologic… | Moderate ecol… | Good ecologic… | Good ecologic… | Good ecologic… | Good e |
| | | 1-1: Pre-HMWB status | High | High | High | High | High | High | High | High | Good | Moderate | Good | Good | High | |
| | | 1-3: Overall ecology | High | High | High | High | High | High | High | High | Good | Moderate | Good | Good | High | |
| | | 1-3-1: Physico-Chem | - | - | - | - | - | - | - | - | - | Good | High | High | High | |
| | | 1-3-1-2: Reactive phosphorus | - | - | - | - | - | - | - | - | - | Good | High | High | High | |
| | | 1-3-2: Biological elements | - | - | - | - | - | - | - | - | Good | Good | Good | Good | - | |
| | | 1-3-2-4: Alien species | - | - | - | - | - | - | - | - | Good | Good | Good | Good | - | |
| | | 1-3-3: Specific pollutants | Pass | Pass | Pass | Pass | Pass | Pass | Pass | Pass | Pass | Fail | Pass | Pass | Pass | |
| | | 1-3-3-14: Ammonium | Pass | Pass | Pass | Pass | Pass | Pass | Pass | Pass | Pass | Fail | Pass | Pass | Pass | |
| | | 4-1: Water quality | High | High | High | High | High | High | High | High | - | - | - | - | - | |

# SEPA Water Classification Hub – Groundwater (bedrock aquifer)

Search information sheets: Search ▶ Return to summary tool

Tool Guidance
Terms and Conditions

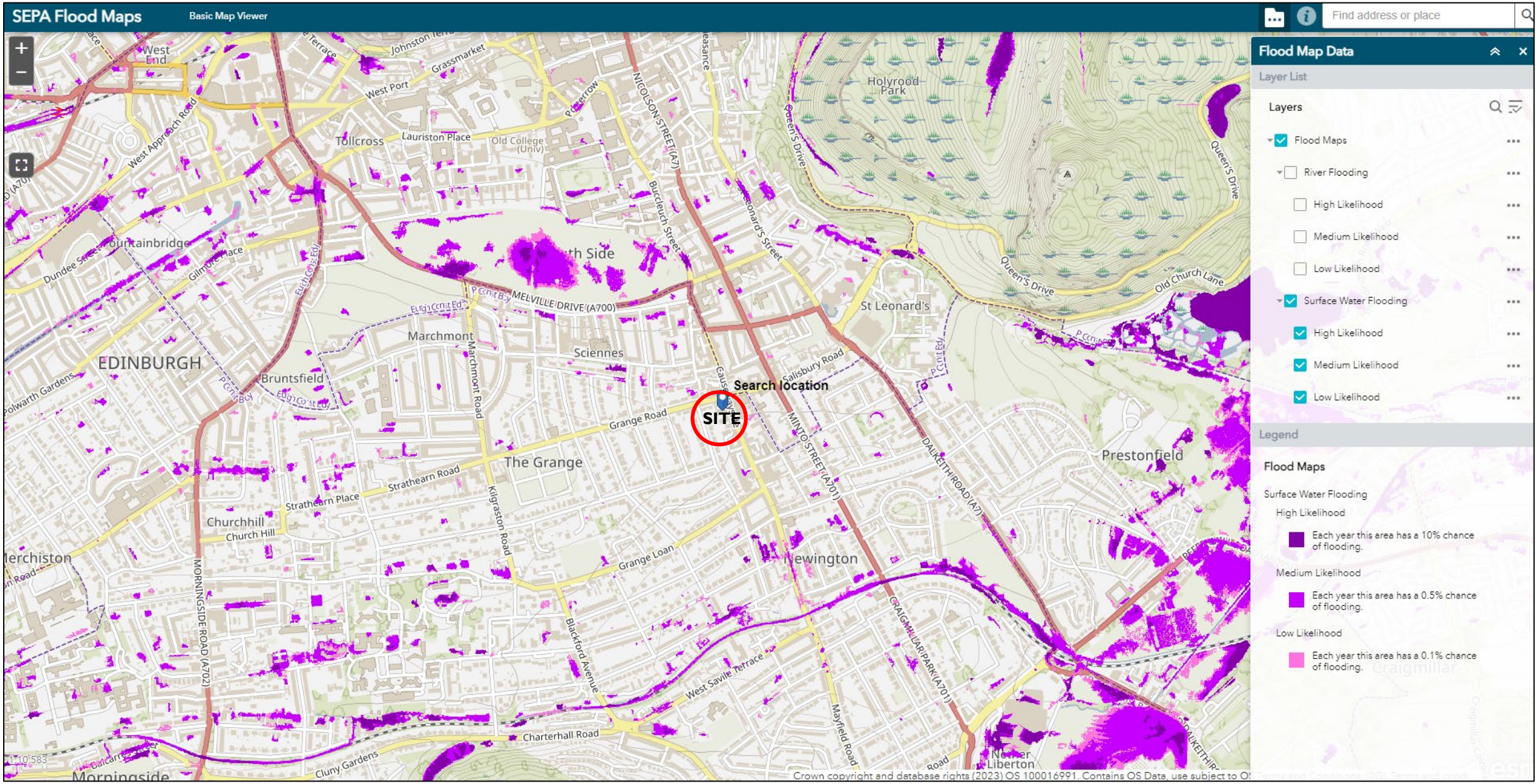
Edinburgh is a groundwater (ID: 150591), in the Scotland river basin district. It is 113.7 square kilometres in area.



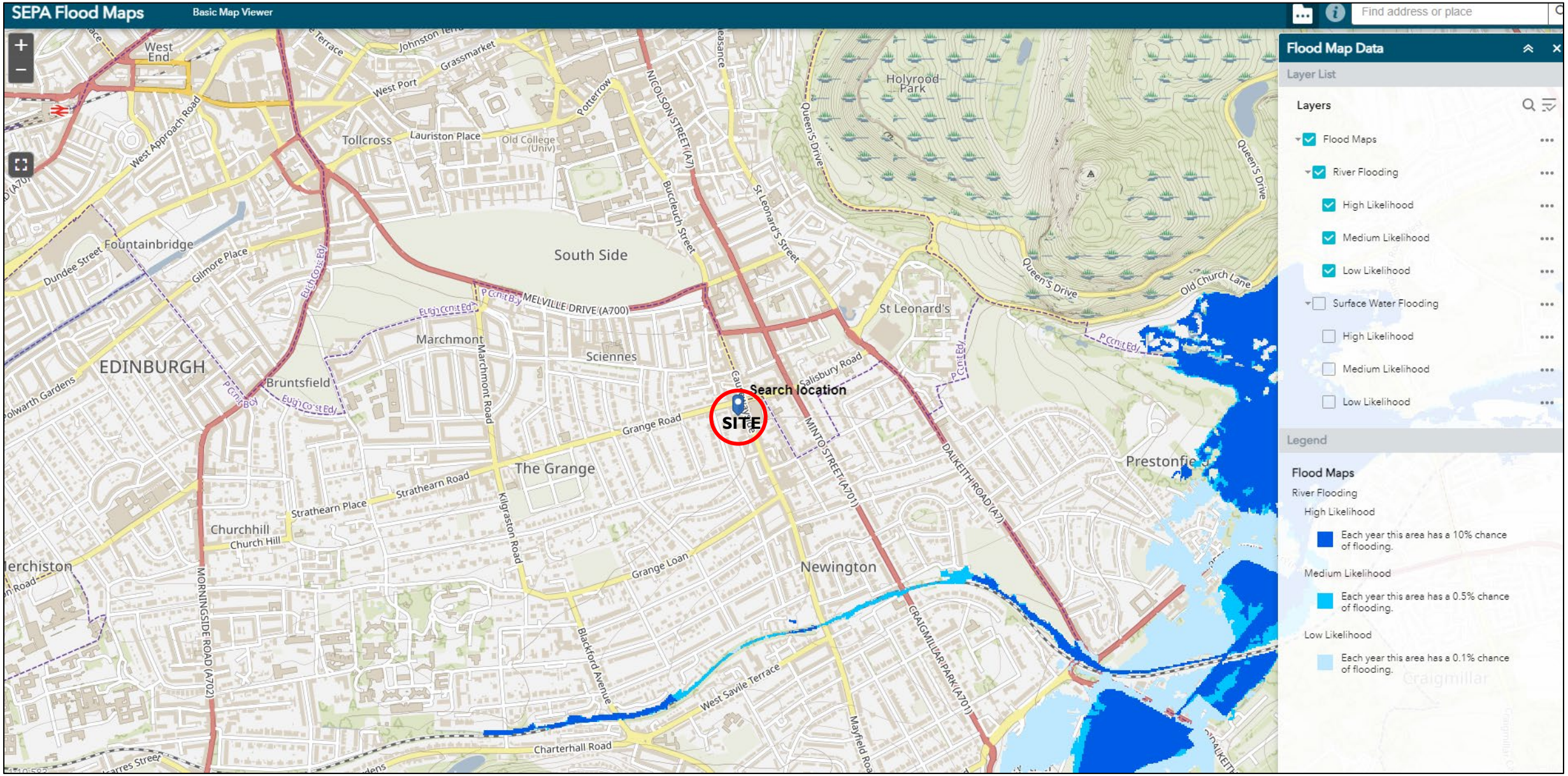

## Water classification data for selected water body

| ID | Name | Parameter | 2020 | 2019 | 2018 | 2017 | 2016 | 2015 | 2014 | 2013 | 2012 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 150591 | Edinburgh | 2: Overall status | Good | Good | Good | Good | Good | Good | Good | Good | Good |
| | | 2-1: Quantitative status | Good | Good | Good | Good | Good | Good | Good | Good | Good |

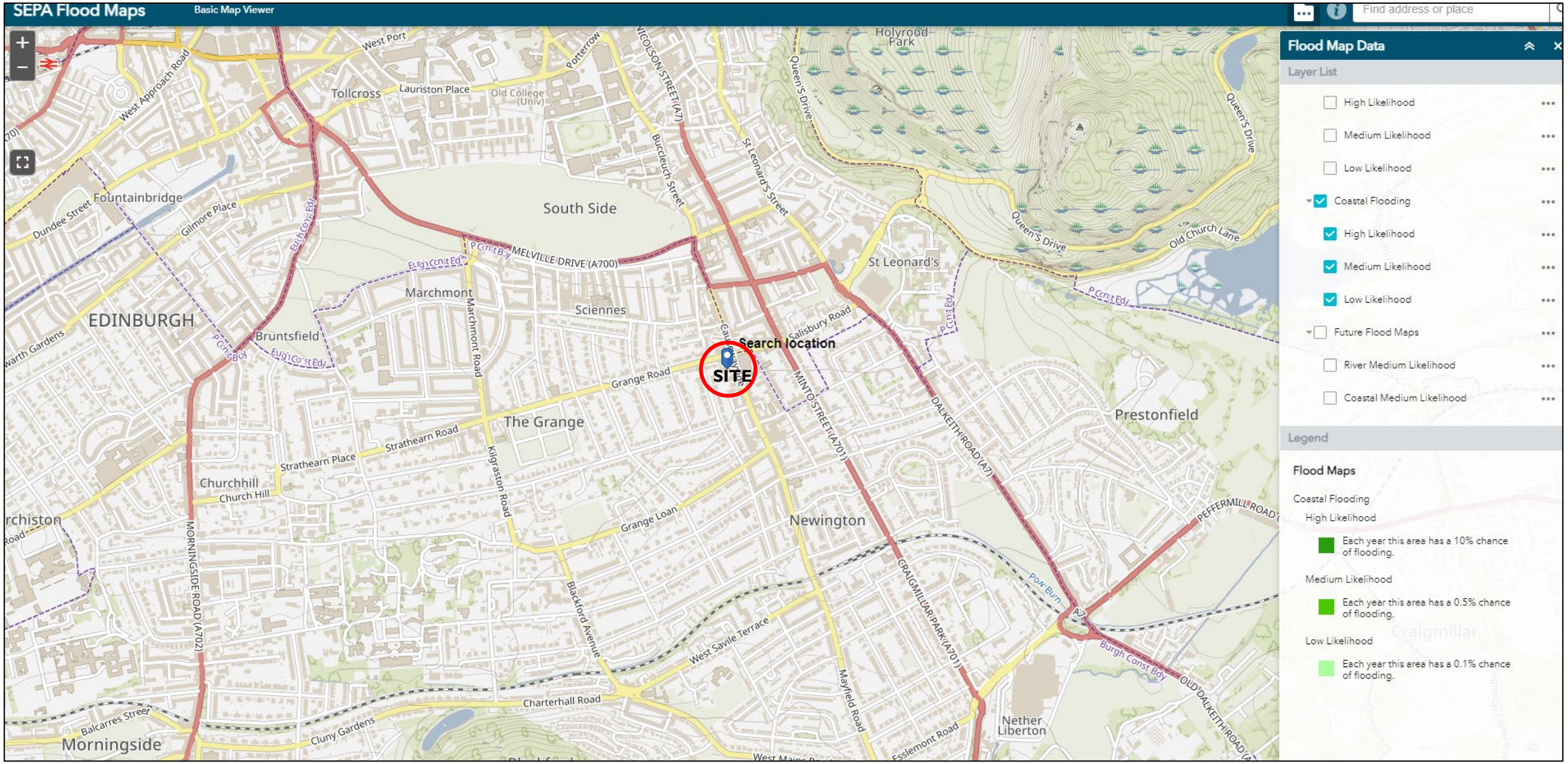
# SEPA Flood Maps – Surface Water Flooding



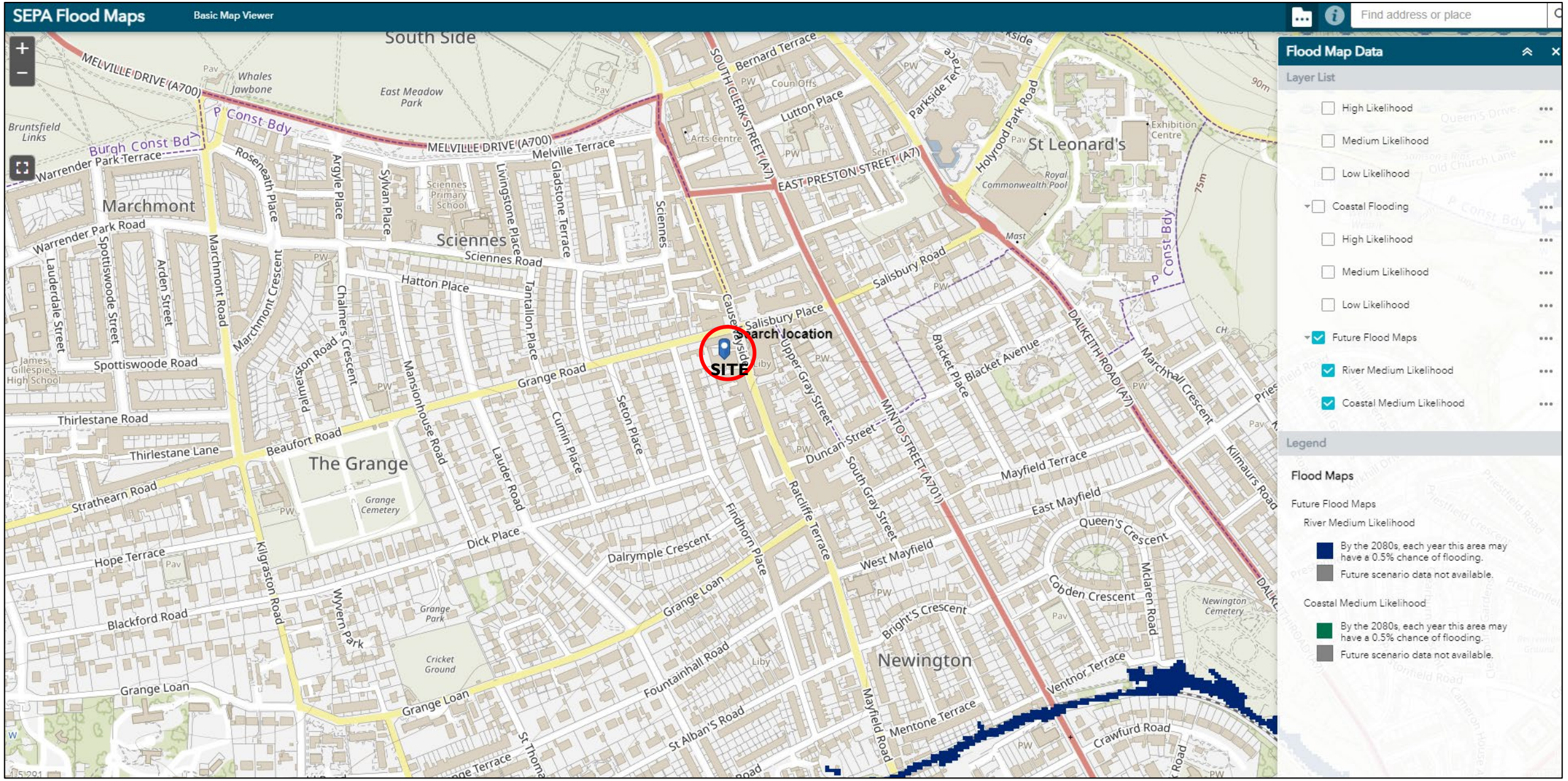
# SEPA Flood Maps – River Flooding



# SEPA Flood Maps – Coastal Flooding



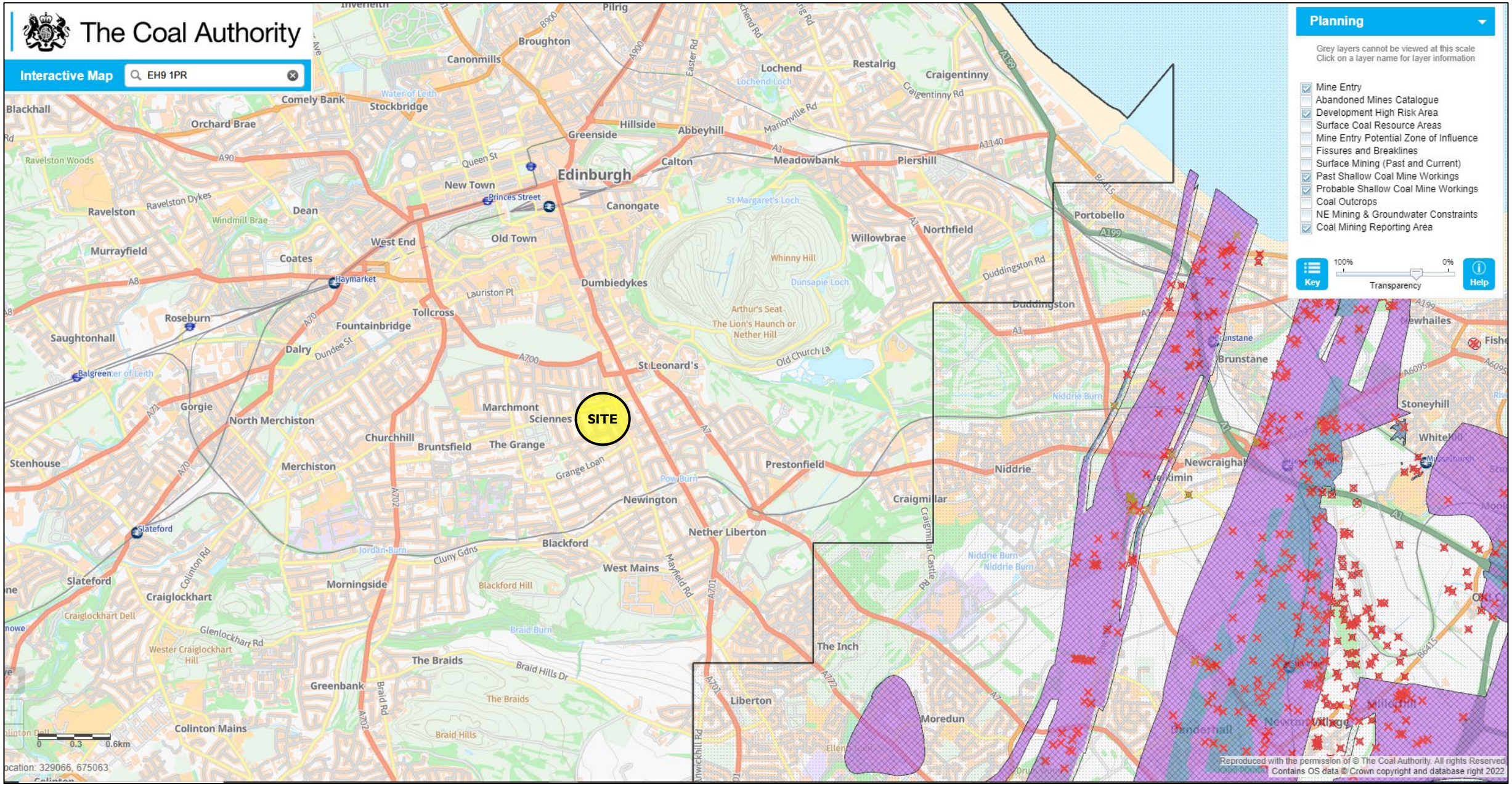
# SEPA Flood Maps – Future Flood Maps (river and coastal)



**Appendix 4**

**The Coal Authority Interactive Viewer Map**

# The Coal Authority Interactive Viewer Map



**Appendix 5**

**BGS Borehole Scans and Data**

British Geological Survey

BGS ID: 1036922 : BGS Reference: NT27SE797/1
EPSG: 27700 : 326408,672158

# RECORD OF BOREHOLE 1

Ground level: 23.10m above O.D. Dia. of boring: 0.20m to 12.10m

Method of boring: Shell and Auger Lining tubes: 0.20m to 11.70m

| Daily Progress | Samples Depth | Samples Type | Change of Strata Legend | Change of Strata Depth | Change of Strata O.D. Level | Description of Strata |
|---|---|---|---|---|---|---|
| | 0.30 | BD | | 0.30 | 22.80 | REINFORCED CONCRETE |
| | 0.60 | BD | | 0.60 | 22.50 | FILL (broken brick) |
| | 1.50-1.95 | U(4) | | 1.35 | 21.75 | FILL (brown sandy clay with fine to coarse gravel and broken brick) |
| 16.5.74 | 1.95 | D | | | | |
| | 2.00 | BD | | | | |
| | 2.15-2.45 | C(93) | | | | |
| | 3.50 | BD | | | | |
| | 3.65-3.95 | C(52) | | | | |
| | 5.00 | BD | | | | |
| | 5.15-5.45 | C(32) | | | | |
| 17.5.74 | | | | | | Stiff brown sandy CLAY with fine to coarse gravel, cobbles and boulders (Boulder Clay) |
| | 6.00 | BD | | | | |
| 18.5.74 | 7.00 | BD | | | | |
| | 7.60-8.05 | U(4) | | | | |
| | 8.05 | D | | | | |
| | 8.55 | D | | | | |
| | 9.55 | BD | | | | |
| 20.5.74 | 10.00-10.45 | U(4) | | | | |
| | 10.45 | D | | | | |
| | 11.70 | D | | | | |
| 21.5.74 | 11.70-11.85 | S(100)* | | | | |
| 22.5.74 | | | | 12.10 | 11.00 | |

Key to type of sample:
U(4) — 102mm (4in.) dia. undisturbed sample
D — disturbed sample
BD — bulk disturbed sample
V — vane test
S ( ) — standard penetration test
C ( ) — dynamic cone penetration test
Figure in brackets is No. of blows for penetration given in depth column (see Notes, page 1).

Remarks: (Observations on ground-water, etc.)

Ground-water was not encountered during boring. Water was added to assist boring below a depth of 3.00m. The borehole was advanced by chiselling between depths of 0.00m to 1.35m (4½h), 2.00m to 7.60m(13½h), 8.10m to 10.00m (5h) and 11.70m to 12.10m (8½h). A standpipe was inserted at a depth of 11.50m in a gravel filter.

* Seating blows only

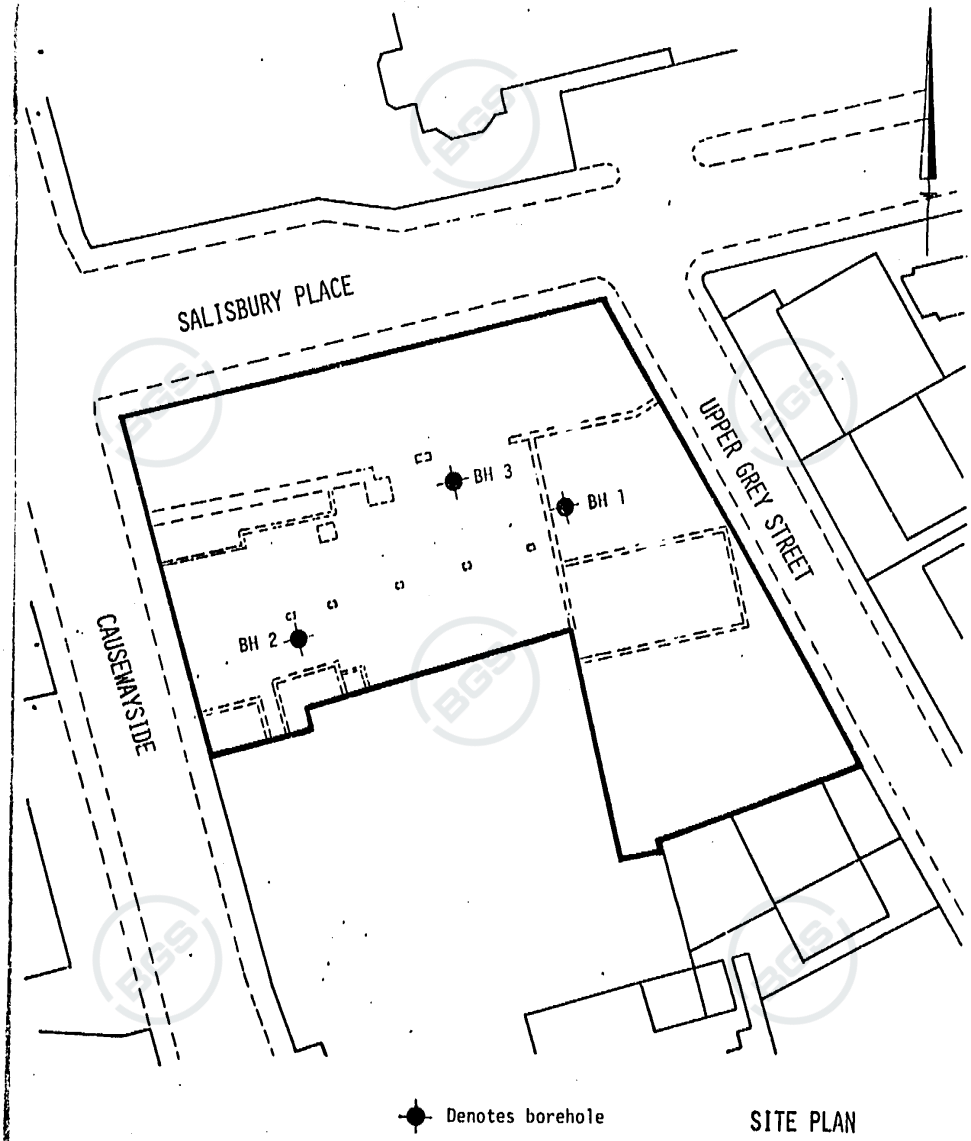
Lab. Ref. No. S/ 10635 | EDINBURGH, CAUSEWAYSIDE | FIG 1

WIMPEY LABORATORIES LIMITED

Contact BGS: ngdc@bgs.ac.uk

British Geological Survey

BGS ID: 1036922 : BGS Reference: NT27SE797/1
EPSG: 27700 : 326408,672158




Denotes borehole

SITE PLAN

SCALE 1:500

| Lab.Ref.No. | | |
|---|---|---|
| S/10635 | EDINBURGH, CAUSEWAYSIDE | FIG 4 |

WIMPEY LABORATORIES LIMITED

Contact BGS: ngdc@bgs.ac.uk

British Geological Survey

BGS ID: 1036924 : BGS Reference: NT27SE797/3
EPSG: 27700 : 326396,672160

# RECORD OF BOREHOLE 3

Ground level: 23.00m above O.D.

Dia. of boring: 0.20m to 11.15m

Method of boring: Shell and Auger

Lining tubes: 0.20m to 2.00m

| Daily Progress | Samples Depth | Samples Type | Change of Strata Legend | Change of Strata Depth | Change of Strata O.D. Level | Description of Strata |
|---|---|---|---|---|---|---|
| | 0.30 | BD | | 0.10 | 22.90 | CONCRETE |
| | 1.00 | BD | | 0.35 | 22.65 | FILL (broken brick) |
| 31.5.74 | | | | 2.10 | 20.90 | FILL (mottled red and brown very sandy CLAY with gravel, brick fragments and occasional boulders) |
| | 2.25-2.70 | U(4) | | | | |
| | 2.70 | D | | | | |
| | 3.75-4.20 | U(4) | | | | |
| | 4.20 | D | | | | Stiff brown sandy CLAY with fine to coarse gravel, cobbles and boulders - boulders encountered at 3.00m, 4.50m, 6.00m and 9.70m |
| 4.6.74 | 5.25-5.70 | U(4) | | | | |
| | 5.70 | D | | | | (Boulder Clay) |
| | 6.75-7.20 | U(4) | | | | |
| | 7.20 | D | | | | |
| | 8.25-8.70 | U(4) | | | | |
| | 8.70 | D | | | | |
| 5.6.74 | | | | | | |
| | 10.15 | BD | | | | |
| | | | | 10.70 | 12.30 | |
| 6.6.74 | | | | 11.15 | 11.85 | BOULDER or ROCK |

Key to type of sample:
U(4) — 102mm (4in.) dia. undisturbed sample
D — disturbed sample
BD — Bulk disturbed sample
V — vane test
S ( ) — standard penetration test
C ( ) — dynamic cone penetration test
Figure in brackets is No. of blows for penetration given in depth column (see Notes, page 1).

Remarks: (Observations on ground-water, etc.)
Ground-water was not encountered during boring. Water was added to assist boring below a depth of 1.80m. The borehole was advanced by chiselling between depths of 1.30m to 1.70m (3½h), 3.00m to 3.55m (3½h), 4.50m to 5.10m (6h), 6.00m to 6.65m (4½h), 9.70m to 10.00m(2½h), and 10.70m to 11.15m(6½h). A standpipe was inserted at a depth of 10.20m in a gravel filter.

Lab. Ref. No. S/10635

EDINBURGH, CAUSEWAYSIDE

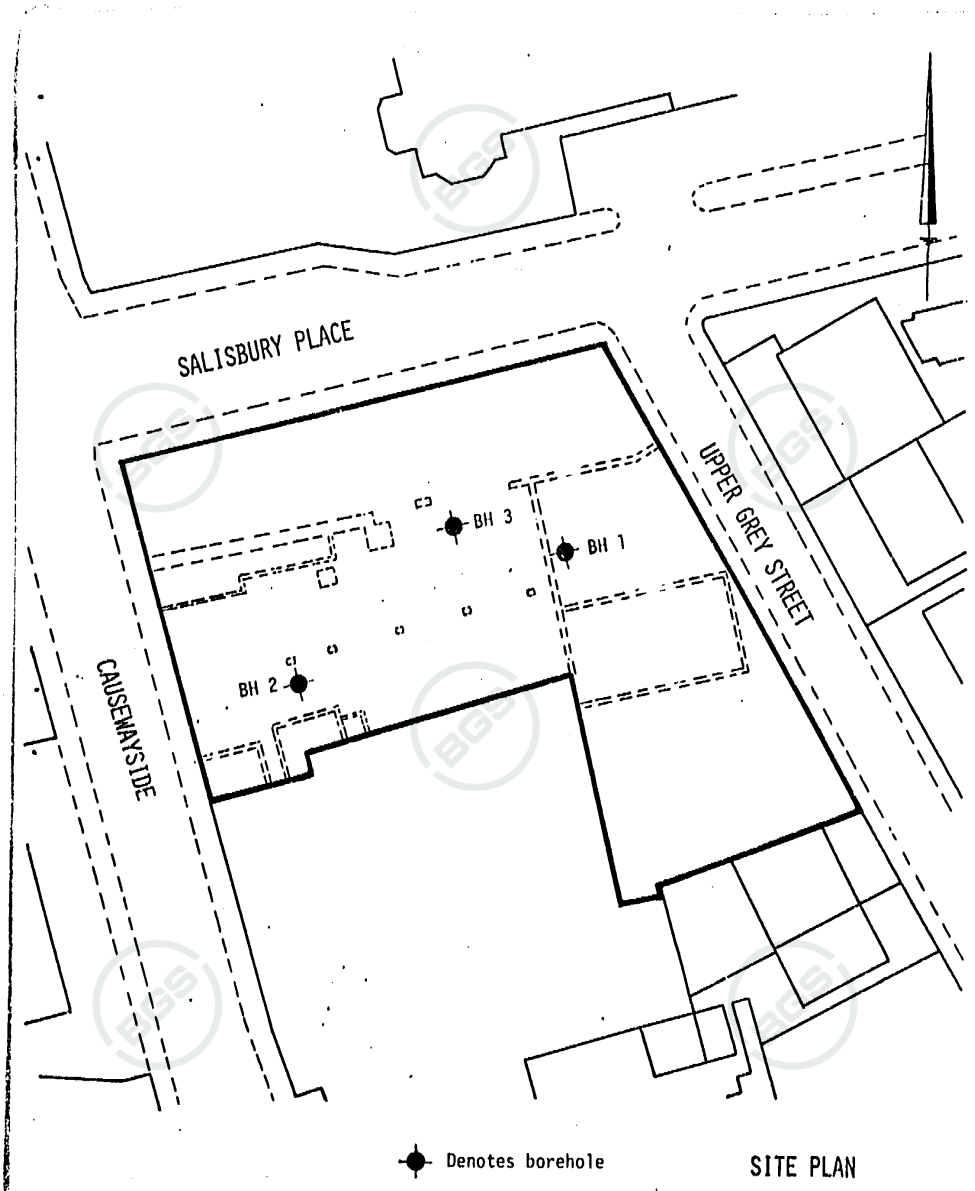
FIG 3

WIMPEY LABORATORIES LIMITED

Contact BGS: ngdc@bgs.ac.uk

British Geological Survey

BGS ID: 1036924 : BGS Reference: NT27SE797/3
EPSG: 27700 : 326396,672160




Denotes borehole

SITE PLAN

SCALE 1:500

| Lab.Ref.No. | | |
|---|---|---|
| S/10635 | EDINBURGH, CAUSEWAYSIDE | FIG 4 |

WIMPEY LABORATORIES LIMITED

Contact BGS: ngdc@bgs.ac.uk

British Geological Survey

BGS ID: 1037297 : BGS Reference: NT27SE1643/A
EPSG: 27700 : 326377,672157

BORING METHOD AND DIAMETER
CABLE TOOL BORING (200 and 150mm)

Dates:- 11 to 17th September 1980.

RECORD OF BOREHOLE A

LOCATION S. 4492 Edinburgh. Fig 1

CLIENT DEPARTMENT OF ENVIRONMENT. Scale 1: 50

| Drilling and Casing Progress | Water Level | C/ϕ | LL/PL | Natural Moisture Content | 'N' Value / R.Q.D. | Sample Core Recovery | Strata Description (Level m (OD) 75.42) | Depth (m) | Legend |
|---|---|---|---|---|---|---|---|---|---|
| 11.9.80 | | | | | | 1 | Concrete | G.L. | |
| | | | | | | 2 | FILL MATERIAL - grey brown speckled silty mixed sand to odd gravel, some broken brick. | 0.20 | |
| | | | | | 17 | 3 | MEDIUM DENSE brown to orange brown clayey very silty fine to medium SAND with odd gravel. | 1.00 | |
| | | | | | | 4 | 1. Almost friable CLAY especially with depth. | | |
| | | 108/0 | 30/13 | 14 | | 5 | STIFF becoming VERY STIFF brown becomes dark brownish grey (below 3m) friable silty sand; CLAY with some mixed GRAVEL to possible boulder(s) | 1.90 | |
| | | | | | | 6 | | | |
| | | | | | | 7 | | | |
| | | | | | 55 | 8 | | | |
| | | | | | | 9 | | | |
| 12.9.80 | | | 28/13 | 11 | | 10 | | | |
| | | | | | | 11 | | | |
| | | | | | 45 | 12 | | | |
| | | | | | | 13 | | | |
| | | | 27/11 | 11 | | 14 | | | |
| | | | | | | 15 | | | |
| | | | | | | 16 | | | |
| | | | | | 46 | 17 | | | |
| 15.9.80 | | | | | | 18 | | | |
| | | | | | | 19 | | 10.00 | |
| | | A10/0 | 25/10 | 10 | | | | | |

REMARKS 1. Chiselling required from G.L. to 1.0m at intervals, where necessary from 2.5 to 12.3m.

KEY W Water; Undisturbed; Disturbed; Bulk disturbed; Water encountered; Standing water level; Drilling Progress; Casing Progress

GKN KELLER FOUNDATIONS

GKNF32

Contact BGS: ngdc@bgs.ac.uk

British Geological Survey

BGS ID: 1037297 : BGS Reference: NT27SE1643/A
EPSG: 27700 : 326377,672157

BORING METHOD AND DIAMETER

CABLE TOOL BORING (200 and 150mm)

Dates:- 11 to 17th September 1980

RECORD OF BOREHOLE A

LOCATION S. 4482, Edinburgh

CLIENT DEPARTMENT OF ENVIRONMENT

Fig 2

Scale 1:50

| Drilling and Casing Progress | Water Level | c/ø | LL/PL | Natural Moisture Content | 'N' Value / R.Q.D. | Sample Core Recovery | Strata Description (Level m (OD) 75.42) | Depth (m) | Legend |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | 20 | VERY STIFF silty sandy CLAY (as before ) | 10.00 | |
| | | | | | | 21 | 2. Sample 24 (jar) contained some brown silty fine SAND. | | |
| | | | | | 70 | 22 | | | |
| | | | | | | 25w | | | |
| | | | | | | 23 | | | |
| | ▽ 12.5.80 | | | | 90 | 24 | | | |
| | | | | | | 26 | | | |
| | | | | | | | END OF BOREHOLE | 12.30 | |

REMARKS 2. Water entered bore at 12.5m and rose to 10.5m after fifteen minutes
3. On morning of 16.9.80, driller remarked "Sand blew up casing to 4m". Water was noted at 3m. Bore was subsequently cleaned out on 16.9.80

KEY W Water; Undisturbed; Disturbed; Bulk disturbed; ▽ Water encountered; ▼ Standing water level; --- Drilling Progress; — Casing Progress

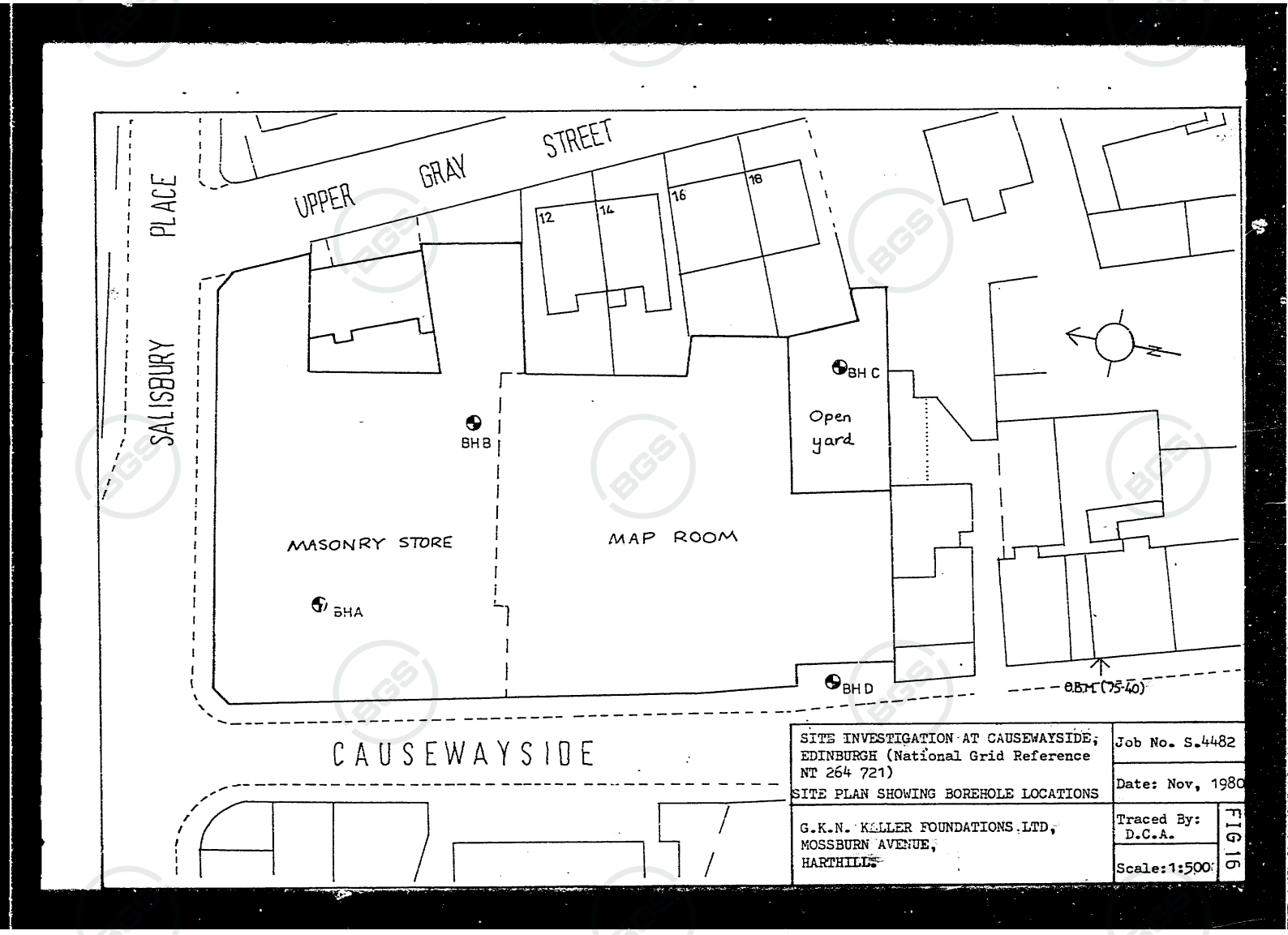
G KELLER FOUNDATIONS

GKNF32

Contact BGS: ngdc@bgs.ac.uk

British Geological Survey

BGS ID: 1037297 : BGS Reference: NT27SE1643/A
EPSG: 27700 : 326377,672157




Contact BGS: ngdc@bgs.ac.uk

British Geological Survey

BGS ID: 1037299 : BGS Reference: NT27SE1643/C
EPSG: 27700 : 326421,672103

BORING METHOD AND DIAMETER

CABLE TOOL BORING (200mm)

Dates:- 2nd to 9th September, 1980

RECORD OF BOREHOLE C

LOCATION S. 4482, Edinburgh. Fig 7

CLIENT DEPARTMENT OF ENVIRONMENT Scale 1: 50

| Drilling and Casing Progress | Water Level | C/ϕ | LL/PL | Natural Moisture Content | 'N' Value / R.Q.D. | Sample / Core / Recovery | Strata Description (Level m (OD) 74.50) | Depth (m) | Legend |
|---|---|---|---|---|---|---|---|---|---|
| 2.9.80 | | | | | | 1 | Concrete | 0.1 | |
| | | | | | | 1 | FILL MATERIAL - brick, concrete etc. in sandy matrix. | 0.20 | |
| | | | | | | 2 | FILL MATERIAL - darkish grey very silty fine sand with odd gravel and concrete (gravel size) traces ash, blaes etc. | | |
| 3.9.80 | | | | | 14 | 3 | 1. Assessed medium dense | | |
| | Seepage | | | | | 4 | Orange brown clayey very silty SAND with some mixed GRAVEL becomes GREY orange mottled silty very sandy CLAY with GRAVEL, some silty clay seams. | 1.4 | |
| | | | | | | 5 | | | |
| | | | 29/14 | 12 | | 6 | STIFF becomes VERY STIFF (although locally, STIFF) initially brown becomes darkish grey (below 2.6m approx.) friable silty sandy CLAY with some mixed GRAVEL to possible boulder(s) | | |
| a.m. 4.9.80 | | | | | | 7 | | | |
| | | | | | | 8 | | | |
| | | | | | | 9 | 1. Gravel, angular to rounded. | | |
| a.m. 4.9.80 | | | | | 51 | 10 | | | |
| 4.9.80 | | | | | | 11 | | | |
| | | | 27/11 | 12 | | 12 | | | |
| | | | | | | 13 | | | |
| 8.9.80 | | | | | | 14 | | | |
| | | | | | 57 | 15 | | | |
| | | | | | | 16 | | | |
| | | | 29/12 | 12 | | 17 | | | |
| | | | | | | 18 | | | |
| | | | | | | 19 | | | |
| | | | | | 96 | 20 | | 10.00 | |

REMARKS
1. Chiselling required from GL to 0.4m (concrete and rubble fill).
2. Undisturbed samples (U-102's) attempted at 1.7, 3.5 and 12m - no penetration.
3. Seepage in bore at 1.7m
4. Chiselling required at intervals from 3.5 to 13.59m.
5. On morning of 4.9.80, water was at 4.0m. Bore sealed dry at end of 4.9.80.
6. On morning of 8.9.80, water had entered bore and settled at 3.0m.

KEY
W Water
Undisturbed
Disturbed
Bulk disturbed
Water encountered
Standing water level
Drilling Progress
Casing Progress

GKN KELLER FOUNDATIONS

GKNF32

Contact BGS: ngdc@bgs.ac.uk

British Geological Survey

BGS ID: 1037299 : BGS Reference: NT27SE1643/C
EPSG: 27700 : 326421,672103

BORING METHOD AND DIAMETER

CABLE TOOL BORING (200mm)

Dates:- 2nd to 9th September 1980.

RECORD OF BOREHOLE C

LOCATION S. 4482, Edinburgh. Fig 8

CLIENT DEPARTMENT OF ENVIRONMENT. Scale 1:50

| Drilling and Casing Progress | Water Level | C/ø | LL/PL | Natural Moisture Content | 'N' Value / R.Q.D. | Sample Core Recovery | Strata Description (Level m (OD) 74.59) | Depth (m) | Legend |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | 21 | VERY STIFF silty sandy CLAY (as before) | 10.00 | |
| 9.9.80 | | | | | | 22 | | | |
| | | | | | | 23 | | | |
| | | | | | 153 | 24 | | | |
| | | | | | | 25 | | | |
| | | | | | | | Bedrock or Boulder - no recovery | 13.50 | |
| | | | | | | | END OF BOREHOLE | 13.59 | |

REMARKS 7. Bore was dry at end of 8.9.80 and on morning of 9.9.80
8. X - SPT at 13.5m - no initial penetration and no recovery possible.

KEY W Water; Undisturbed; Disturbed; Bulk disturbed
Water encountered; Standing water level; --- Drilling Progress; — Casing Progress

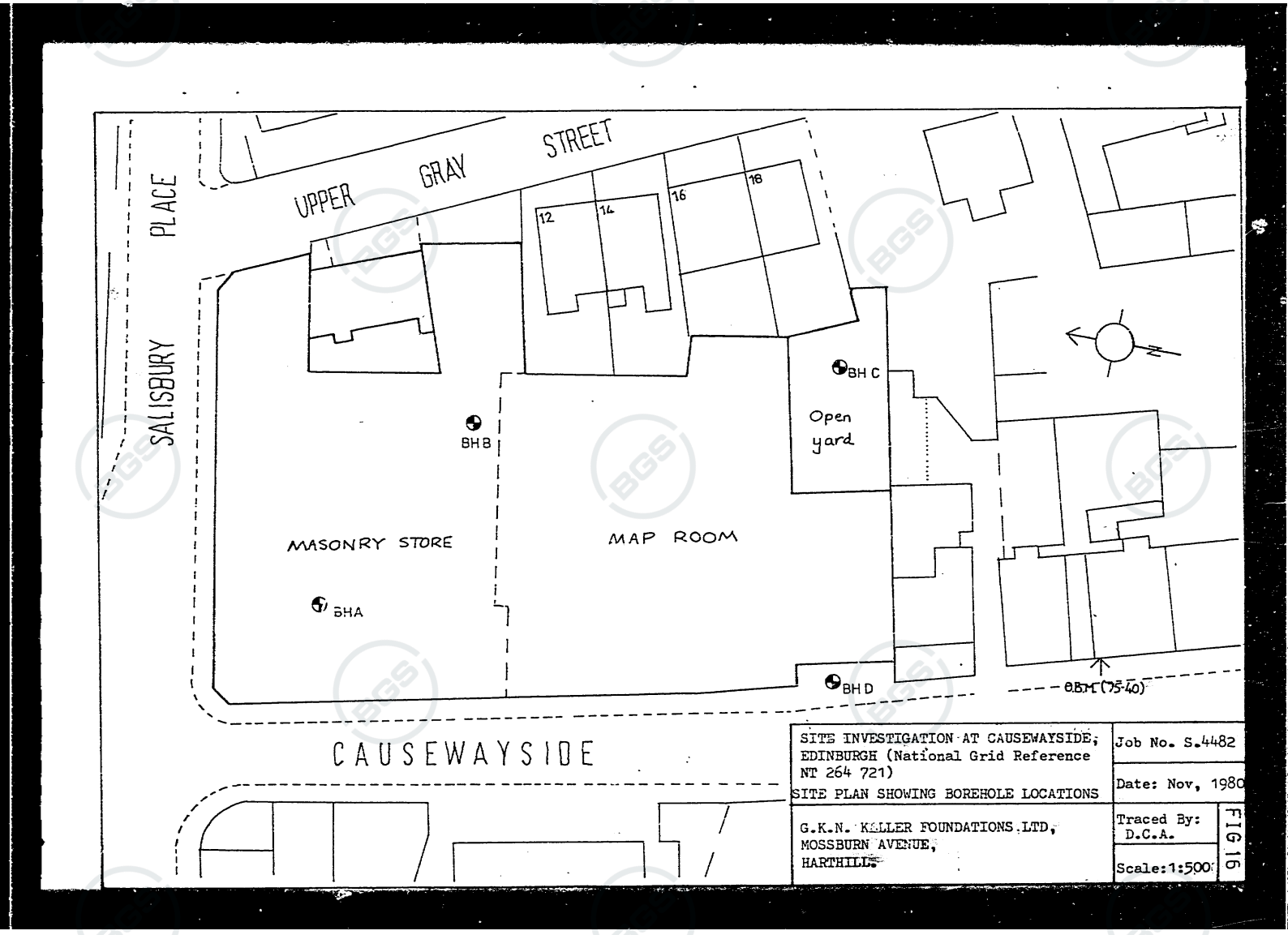
GKN KELLER FOUNDATIONS

GKNF32

Contact BGS: ngdc@bgs.ac.uk

British Geological Survey

BGS ID: 1037299 : BGS Reference: NT27SE1643/C
EPSG: 27700 : 326421,672103




Contact BGS: ngdc@bgs.ac.uk

British Geological Survey

BGS ID: 1037300 : BGS Reference: NT27SE1643/D
EPSG: 27700 : 326384,672093

**BORING METHOD AND DIAMETER**

CABLE TOOL BORING (200 and 250mm) and ROTARY CORE DRILLING (140mm, water flush)

Dates:- 19 to 24th September and 15 to 24th Oct. 1980

**RECORD OF BOREHOLE** D

**LOCATION** S. 4482, Edinburgh. Fig 9

**CLIENT** DEPARTMENT OF ENVIRONMENT. Scale 1: 50

| Drilling and Casing Progress | Water Level | C/ϕ | LL/PL | Natural Moisture Content | 'N' Value / R.Q.D. | Sample / Core / Recovery | Strata Description (Level m (OD) 75.16) | Depth (m) | Legend |
|---|---|---|---|---|---|---|---|---|---|
| 19.9.80 | | | | | | 1 | Tar macadam | GL | |
| | | | | | | 2 | Concrete | 0.15 | |
| | | | | | | 3 | Orange brown clayey very silty sand (almost clay) with broken dolerite gravel inclusions - POSSIBLE FILL. | 0.40 | |
| | | | | | | 4 | | | |
| | | | | | | 5 | Brown to orange brown clayey very silty SAND with odd gravel to cobbles. | 0.70 | |
| | | | 23/11 | | | 6 | | 1.20 | |
| 22.9.80 | | | | | | | STIFF brown, grey tinges friable silty very sandy CLAY with some mixed GRAVEL to odd cobble, some weathered sandstone and coal fragment 1. Boulder indicated from 1.7 to 6.19m. | 2.20 (approx) | |
| 23.9.80 | | | 29/13 | | | 7 | | | |
| | | | | | | 8 | VERY STIFF brown becomes darkish grey (below 2.6m) friable silty sandy CLAY with some GRAVEL to possible boulder(s). | | |
| | | | | | 47 | 9 | | | |
| | | | | | | 10 | 1. Gravel, angular to rounded. | | |
| | | | | | 52 | 11 | | | |
| | | | | | | 12 | | | |
| | | | 28/11 | | | 13 | | | |
| | | | | | 80 | 14 | | | |
| 24.9.80 | p.m. 24.9.80 | | | | | 15 | | | |
| | | | | | 56 | 16 | | | |
| | | | | | | 17 | | | |
| 16.10.80 | | | | | | | | | |
| | | | | | | 93% | | 10.00 | |

REMARKS
1. Hand-excavated pit (for services) from GL to 1m.
2. Undisturbed sample (U-103) attempted at 1m - limited penetration, achieved, material placed in bulk bag.
3. Chiselling from 1.2 to 1.9m. At 1.9m, bore re-positioned to by-pass boulder.
4. Chiselling at intervals from 1.9 to 8m (Undisturbed samples at 2.3 and 5.3m - limited penetration material placed in bulk bags)
5. On morning of 24.9.80, water in bore at 6.5m.

KEY W ● Water; | Undisturbed; ● Disturbed; ♦ Bulk disturbed; ▽ Water encountered; ▼ Standing water level; --- Drilling Progress; — Casing Progress

GKN KELLER FOUNDATIONS

GKNF32

Contact BGS: ngdc@bgs.ac.uk

British Geological Survey

BGS ID: 1037300 : BGS Reference: NT27SE1643/D
EPSG: 27700 : 326384,672093

**BORING METHOD AND DIAMETER**

CABLE TOOL BORING (200 and 250mm) and ROTARY CORE DRILLING (140mm, water flush)

Dates:-19 to 24th Sep. and 15 to 24th Oct. 1980.

**RECORD OF BOREHOLE** D

**LOCATION** S. 4482, Edinburgh. Fig 10

**CLIENT** DEPARTMENT OF ENVIRONMENT. Scale 1:50

| Drilling and Casing Progress | Water Level | C/φ | LL/PL | Natural Moisture Content | 'N' Value / R.Q.D. | Sample Core Recovery | Strata Description (Level m (OD) 25.10) | Depth (m) | Legend |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | 100% | Darkish grey silty sandy CLAY with some GRAVEL to COBBLES<br>2. No boulder size inclusions evident in core sample. | 10.00 | |
| 17.10.80 | | | | | | 24% | | | |
| | | | | | | 38% | | | |
| | | | | | 0% | | VERY WEAK to WEAK highly weathered brown becoming red clayey SILTSTONE (marl type) | 16.30 | |
| 21.10.80 | | | | | 14% | 48% | STRONG to VERY STRONG fresh grey fine grained SANDSTONE.<br>1. Laminated structure and cross-bedding evident, some iron staining. | 16.80 | |
| | | | | | 0% | 97% | MODERATELY STRONG moderately weathered grey clayey SILTSTONE.<br>1. Smooth jointing evident at 18m with iron stained surfaces.<br>2. at 18.5m, rough bedding feature with polished surfaces. | 17.85 | |
| | | | | | | | STRONG fresh grey fine grained SANDSTONE.<br>1. Sub-vertical joint with slight iron-stained surface at 19.3m. | 19.00 | |
| | | | | | 47% | 85% | See over. | 19.60 | |
| | | | | | | | | 20.00 | |

**REMARKS** 4. Rotary core drilling commenced at 8m below ground level.

**KEY**

W Water
Undisturbed
Disturbed
Bulk disturbed
Water encountered
Standing water level
Drilling Progress
Casing Progress

GKN KELLER FOUNDATIONS

GKNF32

Contact BGS: ngdc@bgs.ac.uk

British Geological Survey

BGS ID: 1037300 : BGS Reference: NT27SE1643/D
EPSG: 27700 : 326384,672093

**BORING METHOD AND DIAMETER**

CABLE TOOL BORING (200 and 250mm) and ROTARY CORE DRILLING (140mm, water flush)

Dates:- 19 to 24th Sept. and 15 to 24th Oct. 1980.

**RECORD OF BOREHOLE** D

**LOCATION** S. 4432, Edinburgh. Fig 11

**CLIENT** DEPARTMENT OF ENVIRONMENT. Scale 1:50

| Drilling and Casing Progress | Water Level | C/ϕ | LL/PL | Natural Moisture Content | 'N' Value / R.Q.D. | Sample / Core Recovery | Strata Description (Level m (OD) 75.1?) | Depth (m) | Legend |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | VERY STRONG fresh grey fine grained SANDSTONE. 1. Sub-vertical rough joint from 20.4 to 20.7m- stained and calcite infilled (light joint) | 20.00 | |
| 22.10.80 | | | | | | | STRONG moderately weathered grey and red silty fine SANDSTONE. | 21.30 | |
| | | | | | 0% | 100% | MODERATELY STRONG slightly weathered darkish grey clayey SILTSTONE. 1. With depth, containes frequent thin mudstone laminae and sandstone bands(i.e. band of fine grained sandstone from 23.5 to 23.7m). 2. Frequent badly fragmented zones at top. | 21.95 | |
| 23.10.80 | | | | | 23% | 77% | | | |
| | | | | | 41% | 100% | STRONG to VERY STRONG fresh red fine to medium SANDSTONE. | 24.15 | |
| | | | | | | | END OF BOREHOLE | 25.10 | |

REMARKS 7. Standpipe piezometer installed in borehole at 18.2m.

KEY W Water; Undisturbed; Disturbed; Bulk disturbed; Water encountered; Standing water level; Drilling Progress; Casing Progress

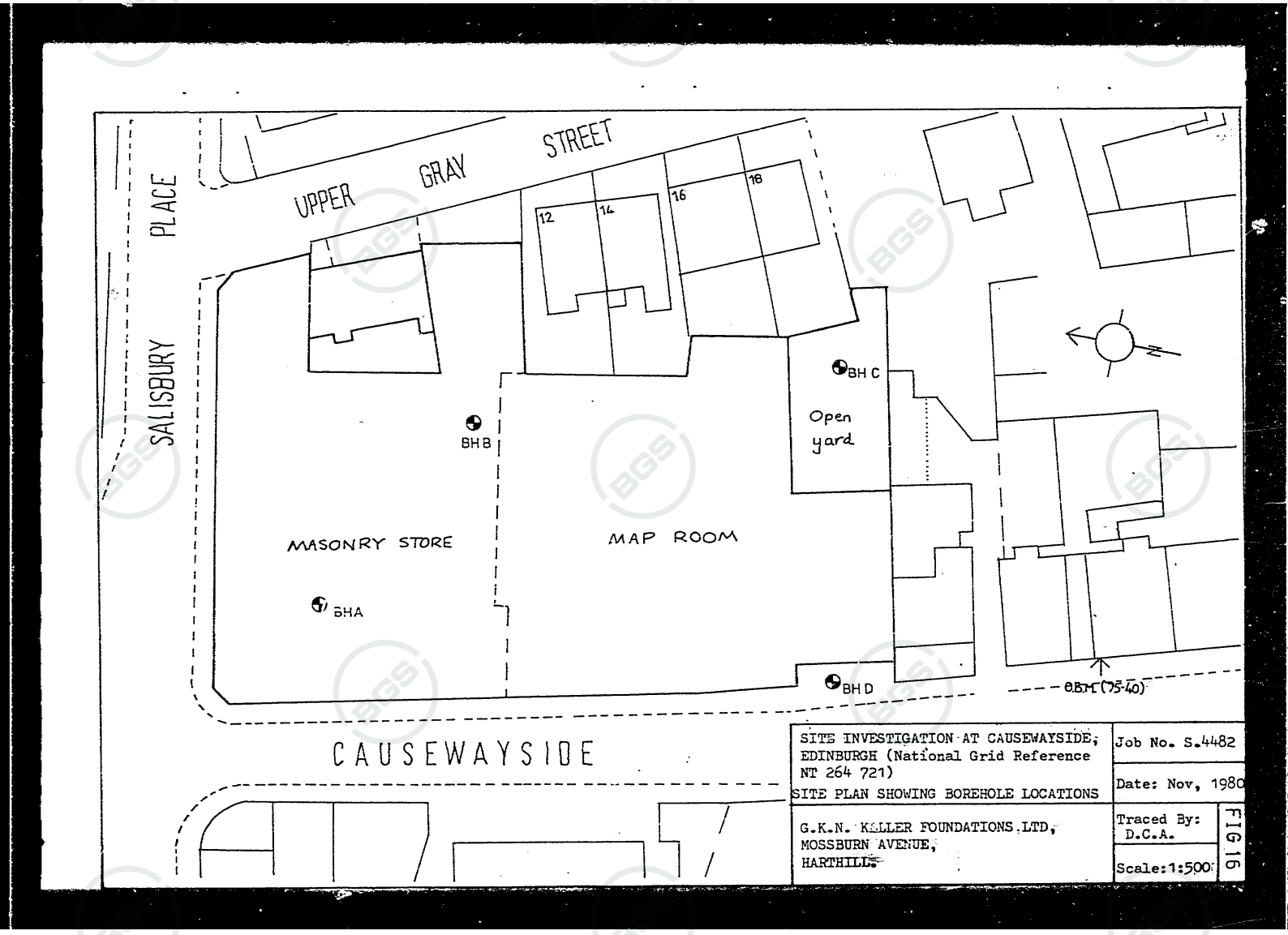
GKN KELLER FOUNDATIONS

GKNF32

Contact BGS: ngdc@bgs.ac.uk

British Geological Survey

BGS ID: 1037300 : BGS Reference: NT27SE1643/D
EPSG: 27700 : 326384,672093




Contact BGS: ngdc@bgs.ac.uk

British Geological Survey

BGS ID: 1037302 : BGS Reference: NT27SE1643/F
EPSG: 27700 : 326403,672143

BORING METHOD AND DIAMETER

CABLE TOOL BORING (200mm)

Dates:- 21st to 23rd July, 1981

RECORD OF BOREHOLE F

LOCATION S.4021, Edinburgh

CLIENT Department of Environment

Fig 3

Scale 1: 50

| Drilling and Casing Progress | Water Level | c/ϕ | LL/PL | Natural Moisture Content | 'N' Value / R.Q.D. | Sample Core Recovery | Strata Description (Level m (OD) 75.45) | Depth (m) | Legend |
|---|---|---|---|---|---|---|---|---|---|
| 21/07/81 | | | | | | 1 | Concrete | 0.10 | |
| | | | | | | 2 | FILL MATERIAL -- broken gravel to cobble size brick, sandstone and concrete fragments with some brown silty sand. | | |
| | | | | | | 3 | | | |
| | | | | | 10 | 5 4 | FILL MATERIAL - 'dirty' brown clayey very sandy silt/very silty sand with frequent small blaes inclusions, traces of broken brick, black ash and sandstone etc, odd orange sandy clay lenses | | |
| | | | | | 6 | 6 | | | |
| | | | | 18 | 6 | 8 7 | FILL MATERIAL - loose orange brown and brown clayey silty sand intermixed with very sandy clay with some gravel traces of blaes and ash | 1.40 | |
| | | | | 18 | 18 | 10 9 | | | |
| | | | | 11 | 23 | 12 11 | SOFT to FIRM orange brown and grey mottled silty sandy CLAY with a little gravel. | 2.50 | |
| | | | | | 28 | 14 13 | SOFT to FIRM becomes STIFF brown grey tinges fissured silty sandy CLAY with a little/some gravel. | | |
| | | | | | | 15 | | | |
| 22/07/81 | ▽ seepage | | | 11 | 40 | 17 16 | VERY STIFF to HARD dark greyish brown friable silty very sandy CLAY with variable GRAVEL to COBBLES and occasional BOULDERS. | | |
| | | | | | 40 | 19 18 | | | |
| | | | | | 40 | 21 20 | 1. Gravel content varies from sub-angular to rounded. | | |
| | ▽ seepage | | | | 32 | 23 22 | 2. Sample No. 24 (jar sample) contained traces of grey very silty fine SAND. | | |
| | | | | | 32 | 25 24 | | | |
| | | | | | 29 | 27 26 | | | |
| | | | | | 41 | 29 28 | | | |
| | | | | | 35 | 31 30 | | | |
| | | | | | 32 | 33 32 | | | |
| | | | | 11 | 32 | 35 34 | | | |
| | | | | | 40 | 37 36 | | | |
| | ▽ | | | | | 38 | | | |
| 23/07/81 | seepage (p.m. 22/07/81) (a.m. 23/07/81) | | | 11 | 33 | 40 39 | | | |
| | | | | | 42 | 42 41 | | | |
| | | | | | | 43 | | | |
| | | | | | 49 | 45 44 | Contd. | 19.35 | |

REMARKS
1. Chiselling required from G.L. to 0.1m (concrete floor)
2. Pick and shovel used from 0.1 to 0.35m.
3. No recovery of SPT sample from 1.15 to 1.60m.
4. Water added to assist bore advancement from 2.05 to 4.05m.
5. Slight water seepage in bore at 4.05m
6. Chiselling required from 3.4 to 4.05m.

KEY W Water; Undisturbed; Disturbed; Bulk disturbed
▽ Water encountered; ▽ Standing water level; Drilling Progress; Casing Progress

GKN KELLER FOUNDATIONS

GKNF32

Contact BGS: ngdc@bgs.ac.uk

British Geological Survey

BGS ID: 1037302 : BGS Reference: NT27SE1643/F
EPSG: 27700 : 326403,672143

BORING METHOD AND DIAMETER

CABLE TOOL BORING (200mm)

Dates:- 21st to 23rd July, 1981

RECORD OF BOREHOLE F contd

LOCATION S.4621, Edinburgh

CLIENT Department of Environment.

Fig 4

Scale 1:50

| Drilling and Casing Progress | Water Level | C/ϕ | LL/PL | Natural Moisture Content | 'N' Value R.Q.D. | Sample Core Recovery | Strata Description (Level m (OD) 75.45) | Depth (m) | Legend |
|---|---|---|---|---|---|---|---|---|---|
| | | for 300mm penetration | | 11 | 91 | 47 46 48 | VERY STIFF to HARD dark greyish brown friable silty very sandy CLAY with variable GRAVEL to COBBLES and occasional BOULDERS. (as before) | 10.50 | |
| | ▽ seepage | | | 8 | 62 | 50 49 | | 11.05 | |
| | | | | | | | END OF BOREHOLE | | |

REMARKS 7. Slight groundwater seepage at 5.25m

8. Chiselling required from 5.85 to 6.3m, 7.65 to 8.1m and 8.3 to 11.05m (total of 4 hours spent chiselling in borehole).

9. At end of 22/07/81 (bore at 8.55m, cased to 7.6m), seepage noted in bore.

10. At start of 23/07/81 seepage noted at base of bore

11. Slight seepage at 11.0m.

12. Casagrande type piezometer installed in bore at 10.5m.

KEY W Water
| Undisturbed
⊘ Disturbed
ϕ Bulk disturbed
▽ Water encountered
▼ Standing water level
--- Drilling Progress
—— Casing Progress

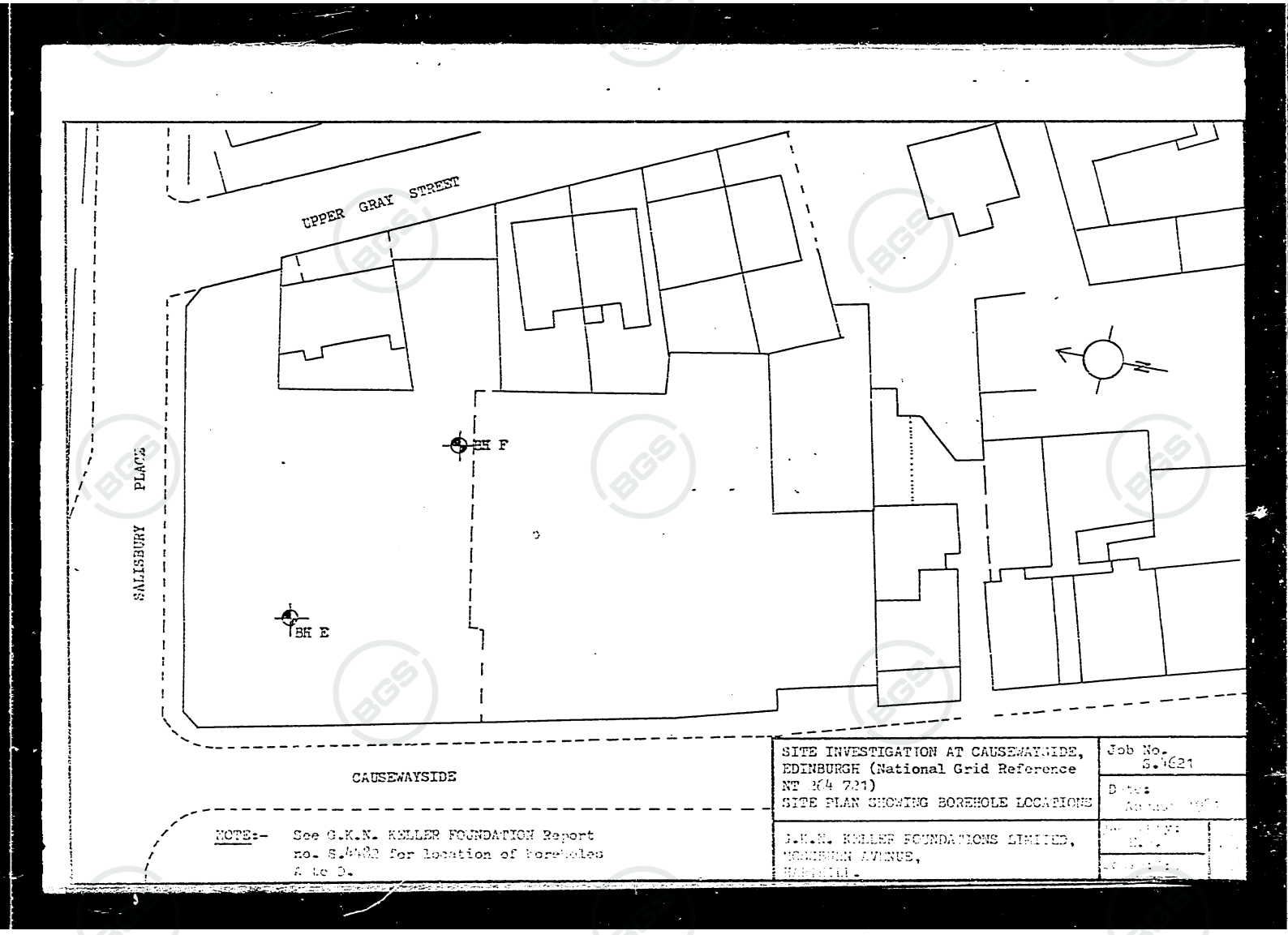
GKN KELLER FOUNDATIONS

GKNF32

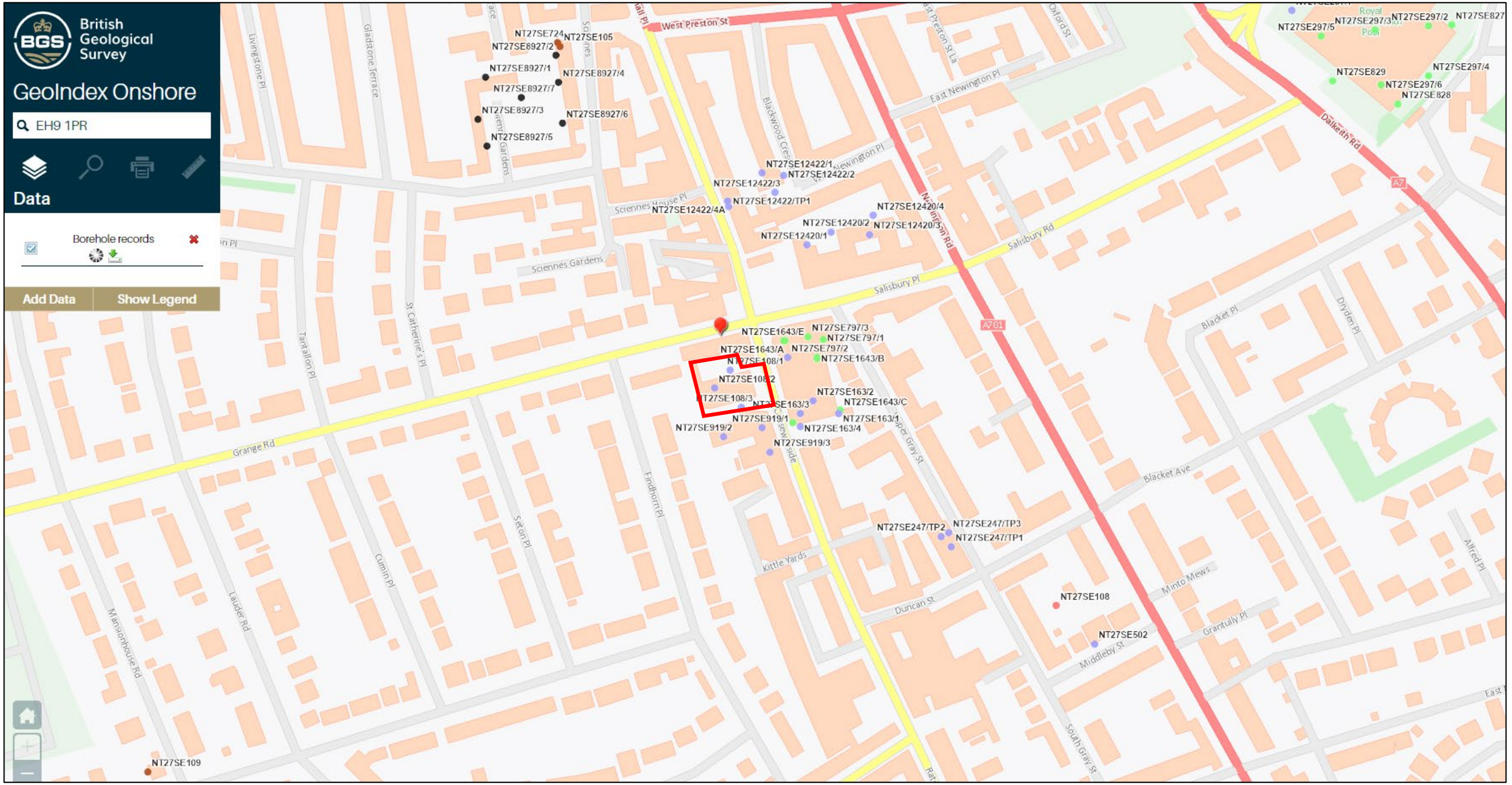
Contact BGS: ngdc@bgs.ac.uk

British Geological Survey

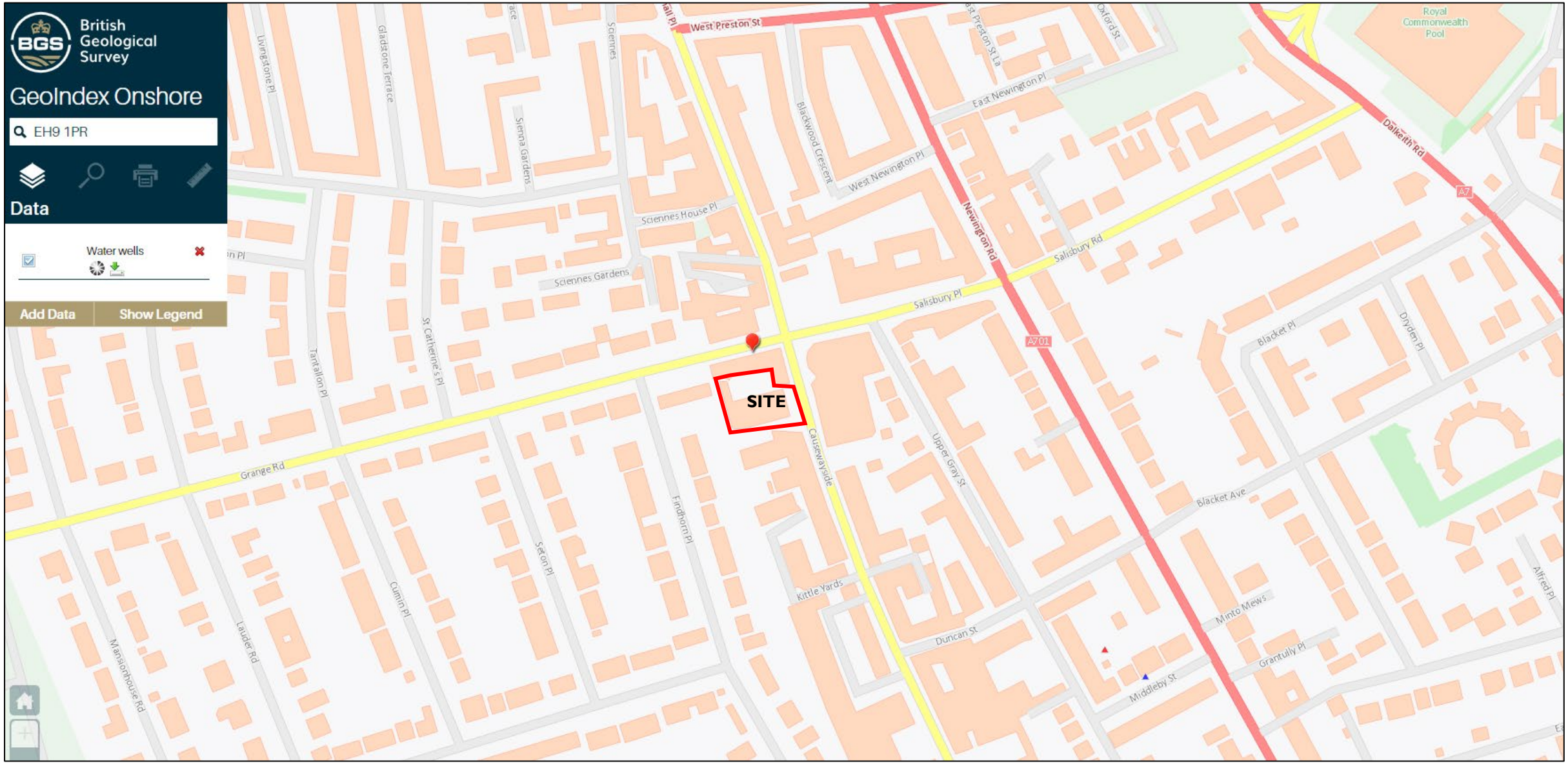
BGS ID: 1037302 : BGS Reference: NT27SE1643/F
EPSG: 27700 : 326403,672143




Contact BGS: ngdc@bgs.ac.uk


British Geological Survey
GeoIndex Onshore
EH9 1PR
Data
Borehole records
Add Data
Show Legend
West Preston St
Salisbury Rd
Salisbury Pl
Sciennes Gardens
Sciennes House Pl
East Newington Pl
Newington Rd
Dalkeith Rd
A7
A701
Grange Rd
Blacket Ave
Blacket Pl
Dryden Pl
Duncan St
Minto Mews
Granfully Pl
Middleby St
Kittle Yards
Findhorn Pl
Cumin Pl
Lauder Rd
Mansionhouse Rd
Seton Pl
St Catherine's Pl
Tantallon Pl
Gladstone Terrace
Livingstone Pl
Blackwood Crescent
Upper Gray St
South Gray St
Alfred Pl
Royal Park
NT27SE724
NT27SE105
NT27SE8927/2
NT27SE8927/1
NT27SE8927/4
NT27SE8927/7
NT27SE8927/3
NT27SE8927/6
NT27SE8927/5
NT27SE12422/1
NT27SE12422/2
NT27SE12422/3
NT27SE12422/TP1
NT27SE12422/4A
NT27SE12420/4
NT27SE12420/2
NT27SE12420/3
NT27SE12420/1
NT27SE1643/E
NT27SE797/3
NT27SE797/1
NT27SE1643/A
NT27SE797/2
NT27SE108/1
NT27SE1643/B
NT27SE108/2
NT27SE108/3
NT27SE163/2
NT27SE163/3
NT27SE1643/C
NT27SE919/1
NT27SE163/1
NT27SE919/2
NT27SE163/4
NT27SE919/3
NT27SE247/TP2
NT27SE247/TP3
NT27SE247/TP1
NT27SE108
NT27SE502
NT27SE109
NT27SE297/5
NT27SE297/3
NT27SE297/2
NT27SE827
NT27SE829
NT27SE297/4
NT27SE297/6
NT27SE828

# BGS GeoIndex – Water Wells



**Appendix 6**

**Records of Exploratory Holes**

# Percussion Drilling Log

Phoenix Drilling Ltd

| | | |
|---|---|---|
| Project Name: Causeway Side | Client: Mason Evans Partnership | Date: 17/04/2023 |
| Location: Edinburgh | Contractor: Phoenix Drilling Limited | |
| Project No. : 4633 | Crew Name: FD | Drilling Equipment: Dando 4000 |

| Borehole Number | Hole Type | Level | Logged By | Scale | Page Number |
|---|---|---|---|---|---|
| BH01 | CP | | CG | 1:50 | Sheet 1 of 1 |

| Well | Water Strikes | Sample and In Situ Testing | | | Depth (m) | Level (m) | Legend | Stratum Description |
|---|---|---|---|---|---|---|---|---|
| | | Depth (m) | Type | Results | | | | |
| | | 0.10 | B | | 0.10 | | | # Tarmacadam (MADE GROUND) |
| | | 0.25 | ES | | | | | Dark blackish brown slightly clayey very gravelly fine to coarse SAND with medium cobble content. Gravel is subangular to angular fine to coarse of sandstone, fine grained igneous, brick and tarmacadam. Cobbles are angular of brick (MADE GROUND). |
| | | 0.50 | B | | 0.50 | | | |
| | | 0.50 | ES | | | | | |
| | | 1.00 | ES | | | | | |
| | | 1.20 | B | | 1.20 | | | Greyish brown slightly silty slightly sandy slightly gravelly CLAY with low cobble content. Sand is fine to coarse. Gravel is subangular to angular fine to coarse of sandstone and fine grained igneous. Cobbles are angular of sandstone and fine grained igneous (MADE GROUND). |
| | | 1.20 | D | | | | | |
| | ▼ | 1.20 | SPT | N=19 (6,4/5,5,4,5) | | | | |
| | ▽ | | | | | | | |
| | | 2.00 | D | | 2.00 | | | Stiff greyish brown slightly silty slightly sandy slightly gravelly CLAY. Sand is fine to coarse. Gravel is subangular to subrounded fine to coarse of sandstone. |
| | | 2.00 | SPT | 50 (25,25/50 for 0mm) | | | | *Terminated on hard strata/obstruction.* |
| | | | | | | | | End of Borehole at 2.000m |

| Hole Diameter | | Casing Diameter | | Chiselling | | | | Inclination and Orientation | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Depth Base | Diameter | Depth Base | Diameter | Depth Top | Depth Base | Duration | Tool | Depth Top | Depth Base | Inclination | Orientation |
| | | | | 2.00 | 2.00 | 01:00 | | | | | |

## Remarks

# Description based on drillers records.
Inspection pit dug to 1.20m and CAT scanned.
SPT refusal and no recovery at 2.00m

# Percussion Drilling Log

Phoenix Drilling Ltd

| | | |
|---|---|---|
| Project Name: Causeway Side | Client: Mason Evans Partnership | Date: 17/04/2023 |
| Location: Edinburgh | Contractor: Phoenix Drilling Limited | |
| Project No. : 4633 | Crew Name: FD | Drilling Equipment: Dando 4000 |

| Borehole Number | Hole Type | Level | Logged By | Scale | Page Number |
|---|---|---|---|---|---|
| BH01A | CP | | CG | 1:50 | Sheet 1 of 1 |

| Well | Water Strikes | Sample and In Situ Testing | | | Depth (m) | Level (m) | Legend | Stratum Description |
|---|---|---|---|---|---|---|---|---|
| | | Depth (m) | Type | Results | | | | |
| | | | | | 0.10 | | | # Tarmacadam (MADE GROUND)<br># Brick ash and stone fill (MADE GROUND). |
| | | | | | 0.50 | | | # Brown sandy CLAY. |
| | | | | | 1.10 | | | Greyish brown slightly silty slightly sandy slightly gravelly CLAY. Sand is fine to coarse. Gravel is subangular to subrounded fine to coarse of sandstone. |
| | ▼ | 2.00 | B | | | | | |
| | | 2.00 | ES | | | | | |
| | | 2.00 | SPT | 50 (25,50/50 for 0mm) | | | | |
| | | 3.00 | B | | | | | |
| | | 3.00 | D | | | | | |
| | | 3.00 | ES | | | | | |
| | | 3.00 | SPT | N=85 (9,13/8,14,13,50) | 3.40 | | | *Terminated on hard strata/obstruction.*<br>End of Borehole at 3.400m |

1 2 3 4 5 6 7 8 9 10

| Hole Diameter | | Casing Diameter | | Chiselling | | | | Inclination and Orientation | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Depth Base | Diameter | Depth Base | Diameter | Depth Top | Depth Base | Duration | Tool | Depth Top | Depth Base | Inclination | Orientation |
| | | | | 3.40 | 3.40 | 00:30 | | | | | |

## Remarks

# Description based on drillers records.
Inspection pit dug to 1.20m and CAT scanned.
SPT refusal and no recovery at 2.00m.

# Percussion Drilling Log

Phoenix Drilling Ltd

| | | |
|---|---|---|
| Project Name: Causeway Side | Client: Mason Evans Partnership | Date: 17/04/2023 |
| Location: Edinburgh | Contractor: Phoenix Drilling Limited | |
| Project No. : 4633 | Crew Name: FD | Drilling Equipment: Dando 4000 |

| Borehole Number | Hole Type | Level | Logged By | Scale | Page Number |
|---|---|---|---|---|---|
| BH01B | CP | | CG | 1:50 | Sheet 1 of 1 |

| Well | Water Strikes | Sample and In Situ Testing | | | Depth (m) | Level (m) | Legend | Stratum Description |
|---|---|---|---|---|---|---|---|---|
| | | Depth (m) | Type | Results | | | | |
| | | | | | 0.10 | | | # Tarmacadam (MADE GROUND)<br># Brick ash and stone fill (MADE GROUND). |
| | | | | | 0.50 | | | # Brown sandy CLAY. |
| | | | | | 1.10 | | | # Brown sandy gravelly CLAY. |
| | | 3.20 | SPT | 50 (25,50/50 for 0mm) | 3.20 | | | *Terminated on hard strata/obstruction.*<br>End of Borehole at 3.200m |

| Hole Diameter | | Casing Diameter | | Chiselling | | | | Inclination and Orientation | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Depth Base | Diameter | Depth Base | Diameter | Depth Top | Depth Base | Duration | Tool | Depth Top | Depth Base | Inclination | Orientation |
| | | | | 3.20 | 3.20 | 00:30 | | | | | |

Remarks

# Description based on drillers records. inspection pit dug to 1.20m and CAT scanned
SPT refusal and no recovery at 3.20m. Borehole left open for rotary follow on.

# Percussion Drilling Log

Phoenix Drilling Ltd

| | | |
|---|---|---|
| Project Name: Causeway Side | Client: Mason Evans Partnership | Date: 18/04/2023 |
| Location: Edinburgh | Contractor: Phoenix Drilling Limited | |
| Project No. : 4633 | Crew Name: FD | Drilling Equipment: Dando 4000 |

| Borehole Number | Hole Type | Level | Logged By | Scale | Page Number |
|---|---|---|---|---|---|
| BH02 | CP | | CG | 1:50 | Sheet 1 of 1 |

| Well | Water Strikes | Sample and In Situ Testing | | | Depth (m) | Level (m) | Legend | Stratum Description |
|---|---|---|---|---|---|---|---|---|
| | | Depth (m) | Type | Results | | | | |
| | | 0.10 | B | | 0.10 | | | # Tarmacadam (MADE GROUND) |
| | | 0.25 | ES | | | | | Dark brownish grey slightly sandy angular cobbles of igneous rock (MADE GROUND. |
| | | 0.40 | B | | 0.40 | | | Dark greyish brown slightly clayey gravelly fine to coarse SAND with common cobble content. Gravel is subangular to angular fine to coarse of sandstone, brick and concrete. Cobbles are angular of brick (MADE GROUND). |
| | | 0.50 | ES | | | | | |
| | | 1.00 | ES | | | | | |
| | | 1.20 | B | | 1.20 | | | Stiff becoming very stiff dark brown sandy gravelly CLAY with high cobble content. Sand is fine to coarse. Gravel is subangular fine to coarse of sandstone and rare brick. Cobbles are subangular of sandstone (MADE GROUND). |
| | | 1.20 | D | | | | | |
| | | 1.20 | SPT | N=19 (5,6/5,5,4,5) | | | | |
| | | 2.00 | B | | | | | |
| | | 2.00 | D | | | | | |
| | | 2.00 | ES | | | | | |
| | | 2.00 | SPT | N=36 (7,8/8,9,9,10) | 2.50 | | | |
| | | 2.50 | B | | | | | Very stiff greyish brown slightly sandy slightly silty slightly gravelly CLAY with low cobble content. Sand is fine to coarse. Gravel is subangular to subrounded fine to coarse of sandstone. Cobbles are subrounded of sandstone. |
| | | 3.00 | B | | | | | |
| | | 3.00 | D | | | | | |
| | | 3.00 | ES | | | | | |
| | | 3.00 | SPT | N=49 (9,11/13,11,10,15) | | | | |
| | | 3.50 | B | | | | | |
| | | 4.00 | SPT | 50 (25,50/50 for 0mm) | 4.00 | | | *Terminated on hard strata/obstruction.* |
| | | | | | | | | End of Borehole at 4.000m |

| Hole Diameter | | Casing Diameter | | Chiselling | | | | Inclination and Orientation | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Depth Base | Diameter | Depth Base | Diameter | Depth Top | Depth Base | Duration | Tool | Depth Top | Depth Base | Inclination | Orientation |
| | | | | 4.00 | 4.00 | 01:00 | | | | | |

## Remarks

# Description based on drillers records.
Inspection pit dug to 1.20m and CAT scanned.
SPT refusal and no recovery at 4.00m.

# Percussion Drilling Log

Phoenix Drilling Ltd

| | | |
|---|---|---|
| Project Name: Causeway Side | Client: Mason Evans Partnership | Date: 18/04/2023 |
| Location: Edinburgh | Contractor: Phoenix Drilling Limited | |
| Project No. : 4633 | Crew Name: FD | Drilling Equipment: Dando 4000 |

| Borehole Number | Hole Type | Level | Logged By | Scale | Page Number |
|---|---|---|---|---|---|
| BH02A | CP | | CG | 1:50 | Sheet 1 of 1 |

| Well | Water Strikes | Sample and In Situ Testing | | | Depth (m) | Level (m) | Legend | Stratum Description |
|---|---|---|---|---|---|---|---|---|
| | | Depth (m) | Type | Results | | | | |
| | | | | | 0.10 | | | # Tarmacadam (MADE GROUND)<br># Stone fill (MADE GROUND). |
| | | | | | 0.40 | | | # Brown clay stone and concrete fill (MADE GROUND). |
| | | | | | 2.40 | | | # Sandy gravelly CLAY. |
| | | 4.00 | SPT | 50 (25,50/50 for 0mm) | 4.00 | | | *Terminated on hard strata/obstruction.*<br>End of Borehole at 4.000m |

| Hole Diameter | | Casing Diameter | | Chiselling | | | | Inclination and Orientation | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Depth Base | Diameter | Depth Base | Diameter | Depth Top | Depth Base | Duration | Tool | Depth Top | Depth Base | Inclination | Orientation |
| | | | | 4.00 | 4.00 | 01:00 | | | | | |

## Remarks

# Description based on drillers records. Inspection pit dug to 1.20m and CAT scanned.
SPT refusal and no recovery at 4.00m. Borehole left open for rotary follow on.

# Percussion Drilling Log

Phoenix Drilling Ltd

| | | |
|---|---|---|
| Project Name: Causeway Side | Client: Mason Evans Partnership | Date: 19/04/2023 |
| Location: Edinburgh | Contractor: Phoenix Drilling Limited | |
| Project No. : 4633 | Crew Name: FD | Drilling Equipment: Dando 4000 |

| Borehole Number | Hole Type | Level | Logged By | Scale | Page Number |
|---|---|---|---|---|---|
| BH03 | CP | | CG | 1:50 | Sheet 1 of 1 |

| Well | Water Strikes | Sample and In Situ Testing | | | Depth (m) | Level (m) | Legend | Stratum Description |
|---|---|---|---|---|---|---|---|---|
| | | Depth (m) | Type | Results | | | | |
| | | 0.10 | B | | 0.10 | | | # Tarmacadam (MADE GROUND) |
| | | 0.25 | ES | | 0.30 | | | Dark brownish grey slightly sandy angular cobbles of igneous rock (MADE GROUND. |
| | | 0.30 | B | | | | | Orange and brown angular cobbles of brick and igneous rock (MADE GROUND). |
| | | 0.50 | ES | | | | | |
| | | 1.00 | ES | | | | | |
| | | 1.20 | B | | 1.20 | | | Hard reddish brown sandy gravelly CLAY with medium cobble content. Sand is fine to coarse. Gravel is subangular to subrounded fine to coarse of sandstone, fine grained igneous and rare brick. Cobbles are subangular of sandstone and fine grained igneous (MADE GROUND). |
| | | 1.20 | D | | | | | |
| | | 1.20 | SPT | 69 (9,11/69 for 225mm) | 1.65 | | | |
| | | 2.00 | ES | | 2.00 | | | # Brick, concrete and stone fill (MADE GROUND). |
| | | 2.00 | SPT | 50 (25,50/50 for 0mm) | | | | *Terminated on hard strata/obstruction.* |
| | | | | | | | | End of Borehole at 2.000m |

| Hole Diameter | | Casing Diameter | | Chiselling | | | | Inclination and Orientation | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Depth Base | Diameter | Depth Base | Diameter | Depth Top | Depth Base | Duration | Tool | Depth Top | Depth Base | Inclination | Orientation |
| | | | | 2.00 | 2.00 | 01:00 | | | | | |

## Remarks

# Description based on drillers records.
Inspection pit dug to 1.20m and CAT scanned.
Borehole left open for rotary follow on.
No recovery SPT at 2.00m.

# Percussion Drilling Log

Phoenix Drilling Ltd

| | | |
|---|---|---|
| Project Name: Causeway Side | Client: Mason Evans Partnership | Date: 20/04/2023 |
| Location: Edinburgh | Contractor: Phoenix Drilling Limited | |
| Project No. : 4633 | Crew Name: FD | Drilling Equipment: Dando 4000 |

| Borehole Number | Hole Type | Level | Logged By | Scale | Page Number |
|---|---|---|---|---|---|
| BH04 | CP | | CG | 1:50 | Sheet 1 of 1 |

| Well | Water Strikes | Sample and In Situ Testing | | | Depth (m) | Level (m) | Legend | Stratum Description |
|---|---|---|---|---|---|---|---|---|
| | | Depth (m) | Type | Results | | | | |
| | | 0.10 | B | | 0.10 | | | # Tarmacadam (MADE GROUND) |
| | | 0.25 | ES | | | | | Dark brownish grey slightly sandy angular cobbles of igneous rock (MADE GROUND. |
| | | 0.50 | ES | | | | | |
| | | | | | 0.65 | | | # Concrete (MADE GROUND). |
| | | | | | 0.70 | | | *Terminated on hard strata/obstruction.* |
| | | | | | | | | End of Borehole at 0.700m |

| Hole Diameter | | Casing Diameter | | Chiselling | | | | Inclination and Orientation | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Depth Base | Diameter | Depth Base | Diameter | Depth Top | Depth Base | Duration | Tool | Depth Top | Depth Base | Inclination | Orientation |
| | | | | | | | | | | | |

Remarks

# Description based on drillers records.
Inspection pit dug to 0.70m and CAT scanned.

# Percussion Drilling Log

Phoenix Drilling Ltd

| | | |
|---|---|---|
| Project Name: Causeway Side | Client: Mason Evans Partnership | Date: 20/04/2023 |
| Location: Edinburgh | Contractor: Phoenix Drilling Limited | |
| Project No. : 4633 | Crew Name: FD | Drilling Equipment: Dando 4000 |

| Borehole Number | Hole Type | Level | Logged By | Scale | Page Number |
|---|---|---|---|---|---|
| BH04A | CP | | CG | 1:50 | Sheet 1 of 1 |

| Well | Water Strikes | Sample and In Situ Testing | | | Depth (m) | Level (m) | Legend | Stratum Description |
|---|---|---|---|---|---|---|---|---|
| | | Depth (m) | Type | Results | | | | |
| | | | | | 0.10 | | | # Tarmacadam (MADE GROUND) |
| | | | | | 0.30 | | | # Stone fill (MADE GROUND). |
| | | | | | | | | *Terminated on hard strata/obstruction.* |
| | | | | | | | | End of Borehole at 0.300m |

| Hole Diameter | | Casing Diameter | | Chiselling | | | | Inclination and Orientation | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Depth Base | Diameter | Depth Base | Diameter | Depth Top | Depth Base | Duration | Tool | Depth Top | Depth Base | Inclination | Orientation |
| | | | | | | | | | | | |

Remarks

# Description based on drillers records.
Inspection pit dug to 0.30m and CAT scanned.

# Percussion Drilling Log

Phoenix Drilling Ltd

| | | |
|---|---|---|
| Project Name: Causeway Side | Client: Mason Evans Partnership | Date: 20/04/2023 |
| Location: Edinburgh | Contractor: Phoenix Drilling Limited | |
| Project No. : 4633 | Crew Name: FD | Drilling Equipment: Dando 4000 |

| Borehole Number | Hole Type | Level | Logged By | Scale | Page Number |
|---|---|---|---|---|---|
| BH04B | CP | | CG | 1:50 | Sheet 1 of 1 |

| Well | Water Strikes | Sample and In Situ Testing | | | Depth (m) | Level (m) | Legend | Stratum Description |
|---|---|---|---|---|---|---|---|---|
| | | Depth (m) | Type | Results | | | | |
| | | | | | 0.10 | | | # Tarmacadam (MADE GROUND) |
| | | | | | 0.30 | | | # Stone fill (MADE GROUND). |
| | | | | | | | | *Terminated on hard strata/obstruction.* |
| | | | | | | | | End of Borehole at 0.300m |

| Hole Diameter | | Casing Diameter | | Chiselling | | | | Inclination and Orientation | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Depth Base | Diameter | Depth Base | Diameter | Depth Top | Depth Base | Duration | Tool | Depth Top | Depth Base | Inclination | Orientation |
| | | | | | | | | | | | |

Remarks

# Description based on drillers records.
Inspection pit dug to 0.30m and CAT scanned.

# Percussion Drilling Log

Phoenix Drilling Ltd

| | | |
|---|---|---|
| Project Name: Causeway Side | Client: Mason Evans Partnership | Date: 20/04/2023 |
| Location: Edinburgh | Contractor: Phoenix Drilling Limited | |
| Project No. : 4633 | Crew Name: FD | Drilling Equipment: Dando 4000 |

| Borehole Number | Hole Type | Level | Logged By | Scale | Page Number |
|---|---|---|---|---|---|
| BH05 | CP | | CG | 1:50 | Sheet 1 of 1 |

| Well | Water Strikes | Sample and In Situ Testing | | | Depth (m) | Level (m) | Legend | Stratum Description |
|---|---|---|---|---|---|---|---|---|
| | | Depth (m) | Type | Results | | | | |
| | | 0.10 | B | | 0.10 | | | # Tarmacadam (MADE GROUND) |
| | | 0.25 | ES | | 0.30 | | | # Stone fill (MADE GROUND). |
| | | | | | | | | *Terminated on hard strata/obstruction.* |
| | | | | | | | | End of Borehole at 0.300m |

1 2 3 4 5 6 7 8 9 10

| Hole Diameter | | Casing Diameter | | Chiselling | | | | Inclination and Orientation | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Depth Base | Diameter | Depth Base | Diameter | Depth Top | Depth Base | Duration | Tool | Depth Top | Depth Base | Inclination | Orientation |
| | | | | | | | | | | | |

Remarks

# Description based on drillers records.
Inspection pit dug to 0.30m and CAT scanned.

# Percussion Drilling Log

Phoenix Drilling Ltd

| | | |
|---|---|---|
| Project Name: Causeway Side | Client: Mason Evans Partnership | Date: 20/04/2023 |
| Location: Edinburgh | Contractor: Phoenix Drilling Limited | |
| Project No. : 4633 | Crew Name: FD | Drilling Equipment: Dando 4000 |

| Borehole Number | Hole Type | Level | Logged By | Scale | Page Number |
|---|---|---|---|---|---|
| BH05A | CP | | CG | 1:50 | Sheet 1 of 1 |

| Well | Water Strikes | Sample and In Situ Testing | | | Depth (m) | Level (m) | Legend | Stratum Description |
|---|---|---|---|---|---|---|---|---|
| | | Depth (m) | Type | Results | | | | |
| | | 0.10 | B | | 0.10 | | | # Tarmacadam (MADE GROUND) |
| | | 0.25 | ES | | | | | |
| | | 0.50 | B | | 0.50 | | | Dark brownish grey slightly sandy angular cobbles of igneous rock (MADE GROUND. |
| | | 0.50 | ES | | | | | |
| | | 1.00 | ES | | | | | Stiff greyish brown sandy gravelly CLAY with low cobble content. Sand is fine to coarse. Gravel is subangular to subrounded fine to coarse of sandstone, fine grained igneous and rare brick. Cobbles are subangular of sandstone and fine grained igneous (MADE GROUND). |
| | | 1.20 | D | | | | | |
| | | 1.20 - 1.90 | U | | | | | |
| | | 1.20 | SPT | N=24 (5,6/6,5,7,6) | 1.50 | | | |
| | | 1.90 | SPT | 50 (25,50/50 for 0mm) | 1.90 | | | Stiff greyish brown slightly silty slightly sandy slightly gravelly CLAY. Sand is fine to coarse. Gravel is subangular to subrounded fine to coarse of sandstone. |
| | | | | | | | | End of Borehole at 1.900m |

| Hole Diameter | | Casing Diameter | | Chiselling | | | | Inclination and Orientation | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Depth Base | Diameter | Depth Base | Diameter | Depth Top | Depth Base | Duration | Tool | Depth Top | Depth Base | Inclination | Orientation |
| | | | | | | | | | | | |

## Remarks

# Description based on drillers records.
Inspection pit dug to 1.20m and CAT scanned.
No recovery SPT at 1.90m.

# Percussion Drilling Log

Phoenix Drilling Ltd

| | | |
|---|---|---|
| Project Name: Causeway Side | Client: Mason Evans Partnership | Date: 20/04/2023 |
| Location: Edinburgh | Contractor: Phoenix Drilling Limited | |
| Project No. : 4633 | Crew Name: FD | Drilling Equipment: Dando 4000 |

| Borehole Number | Hole Type | Level | Logged By | Scale | Page Number |
|---|---|---|---|---|---|
| BH06 | CP | | CG | 1:50 | Sheet 1 of 1 |

| Well | Water Strikes | Sample and In Situ Testing | | | Depth (m) | Level (m) | Legend | Stratum Description |
|---|---|---|---|---|---|---|---|---|
| | | Depth (m) | Type | Results | | | | |
| | | 0.10 | B | | 0.10 | | | # Tarmacadam (MADE GROUND) |
| | | 0.25 | ES | | 0.30 | | | Dark brownish grey slightly sandy angular cobbles of igneous rock (MADE GROUND. |
| | | 0.30 | B | | | | | Greyish brown sandy angular fine to coarse gravel of crushed igneous rock (MADE GROUND). |
| | | 0.50 | ES | | | | | |
| | | 1.00 | ES | | | | | |
| | | 1.20 | D | | 1.20 | | | Very stiff greyish brown slightly clayey gravelly fine to coarse SAND with medium cobble content and pockets of brown clay. Sand is fine to coarse. Gravel is subangular fine to coarse of sandstone, fine grained igneous and rare concrete. Cobbles are subangular of fine grained igneous (MADE GROUND) |
| | | 1.20 - 1.70 | U | | | | | |
| | | 1.20 | SPT | N=32 (7,8/8,7,9,8) | | | | |
| | | 1.70 | SPT | 50 (25,50/50 for 0mm) | 1.70 | | | |
| | | | | | | | | *Terminated on hard strata/obstruction.* |
| | | | | | | | | End of Borehole at 1.700m |

| Hole Diameter | | Casing Diameter | | Chiselling | | | | Inclination and Orientation | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Depth Base | Diameter | Depth Base | Diameter | Depth Top | Depth Base | Duration | Tool | Depth Top | Depth Base | Inclination | Orientation |
| | | | | | | | | | | | |

Remarks

# Description based on drillers records.
Inspection pit dug to 1.20m and CAT scanned.
No recovery SPT at 1.70m.

# Rotary Core Log

Phoenix Drilling Ltd

| | | |
|---|---|---|
| Project Name: Causeway Side | Client: Mason Evans Partnership | Date: 20/04/2023 |
| Location: Edinburgh | Contractor: Phoenix Drilling Limited | |
| Project No. : 4633 | Crew Name: JH | Drilling Equipment: Fraste XL |

| Borehole Number | Hole Type | Level | Logged By | Scale | Page Number |
|---|---|---|---|---|---|
| RBH01B | RC | | CG | 1:50 | Sheet 1 of 3 |

| Well | Water | Depth (m) | Type /FI | Coring TCR | Coring SCR | Coring RQD | Diameter Recovery (SPT) | Depth (m) | Level (m) | Legend | Stratum Description |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | Carried out by cable percussion rig. |
| | | 3.50 | SPT | | | | (50) | 3.40<br>3.50 | | | # Boulders. |
| | | 5.00 | SPT | | | | (50) | | | | Very stiff dark brown sandy very gravelly CLAY with high cobble and medium boulder content. Sand is fine to coarse. Gravel is subangular to subrounded fine to coarse of sandstone, quartz and mudstone. Cobbles and boulders are subangular to subrounded of sandstone. |
| | | 6.50 | SPT | | | | (34) | | | | Very stiff dark brown sandy very gravelly CLAY with high cobble and medium boulder content. Sand is fine to coarse. Gravel is subangular to subrounded fine to coarse of sandstone, quartz and mudstone. Cobbles and boulders are subangular to subrounded of sandstone. |
| | | 8.00 | SPT | | | | (34) | | | | |
| | | 9.50 | SPT | | | | (45) | | | | |
| | | | Type/FI | TCR | SCR | RQD | D/R/(SPT) | | | | |

1 2 3 4 5 6 7 8 9 10

| Hole Diameter | | Casing Diameter | | Chiselling | | | | Inclination and Orientation | | | | Drilling Flush | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Depth Base | Diameter | Depth Base | Diameter | Depth Top | Depth Base | Duration | Tool | Depth Top | Depth Base | Inclination | Orientation | Depth Top | Depth Base | Type | Colour | Min (%) | Max (%) |
| 16.00<br>21.00 | 139<br>115 | 16.00 | 139 | | | | | | | | | 0.00 | 21.00 | Water | | | 100 |

Remarks

Borehole advanced to 21.0m using rotary coring drilling techniques to collect rock core samples and terminated on target depth. Borehole fitted with wellpoint upon completion. # Description based on drillers records. FI of 25 = Non-Intact.

# Rotary Core Log

Phoenix Drilling Ltd

| | | |
|---|---|---|
| Project Name: Causeway Side | Client: Mason Evans Partnership | Date: 20/04/2023 |
| Location: Edinburgh | Contractor: Phoenix Drilling Limited | |
| Project No. : 4633 | Crew Name: JH | Drilling Equipment: Fraste XL |

| Borehole Number | Hole Type | Level | Logged By | Scale | Page Number |
|---|---|---|---|---|---|
| RBH01B | RC | | CG | 1:50 | Sheet 2 of 3 |

| Well | Water | Depth (m) | Type /FI | Coring TCR | Coring SCR | Coring RQD | Diameter Recovery (SPT) | Depth (m) | Level (m) | Legend | Stratum Description |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 11.00 | SPT | | | | (48) | | | | Very stiff dark brown sandy very gravelly CLAY with high cobble and medium boulder content. Sand is fine to coarse. Gravel is subangular to subrounded fine to coarse of sandstone, quartz and mudstone. Cobbles and boulders are subangular to subrounded of sandstone. |
| | | 12.50 | SPT | | | | (52) | | | | |
| | | 14.00 | SPT | | | | (47) | | | | |
| | | 15.50 | SPT | | | | (50) | 15.70 | | | |
| | | 16.00 - 18.00 | 25 | 80 | 0 | 0 | | 15.70 | | | Very weak locally extremely weak light grey thinly to thickly bedded micaceous MUDSTONE. Recovered predominantly non-intact and moderately weathered throughout with locally heavily weathered sections. Intact sections fractures are 0-90 degrees closely spaced undulating rough with some staining and infilling from weathering. |
| | | 18.00 - 21.00 | 25<br>5<br>25<br>3<br>25<br>3 | 100 | 51 | 20 | | | | | |
| | | | Type/FI | TCR | SCR | RQD | D/R/(SPT) | | | | |

| Hole Diameter | | Casing Diameter | | Chiselling | | | | Inclination and Orientation | | | | Drilling Flush | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Depth Base | Diameter | Depth Base | Diameter | Depth Top | Depth Base | Duration | Tool | Depth Top | Depth Base | Inclination | Orientation | Depth Top | Depth Base | Type | Colour | Min (%) | Max (%) |
| 16.00<br>21.00 | 139<br>115 | 16.00 | 139 | | | | | | | | | 0.00 | 21.00 | Water | | | 100 |

Remarks

Borehole advanced to 21.0m using rotary coring drilling techniques to collect rock core samples and terminated on target depth. Borehole fitted with wellpoint upon completion. # Description based on drillers records. FI of 25 = Non-Intact.

# Rotary Core Log

Phoenix Drilling Ltd

| | | |
|---|---|---|
| Project Name: Causeway Side | Client: Mason Evans Partnership | Date: 20/04/2023 |
| Location: Edinburgh | Contractor: Phoenix Drilling Limited | |
| Project No. : 4633 | Crew Name: JH | Drilling Equipment: Fraste XL |

| Borehole Number | Hole Type | Level | Logged By | Scale | Page Number |
|---|---|---|---|---|---|
| RBH01B | RC | | CG | 1:50 | Sheet 3 of 3 |

| Well | Water | Depth (m) | Type /FI | Coring TCR | Coring SCR | Coring RQD | Diameter Recovery (SPT) | Depth (m) | Level (m) | Legend | Stratum Description |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 25 | | | | | | | | Very weak locally extremely weak light grey thinly to thickly bedded micaceous MUDSTONE. Recovered predominantly non-intact and moderately weathered throughout with locally heavily weathered sections. Intact sections fractures are 0-90 degrees closely spaced undulating rough with some staining and infilling from weathering. |
| | | | 8 | | | | | 21.00 | | | End of Borehole at 21.000m |
| | | | Type/FI | TCR | SCR | RQD | D/R/(SPT) | | | | |

21 22 23 24 25 26 27 28 29 30

| Hole Diameter | | Casing Diameter | | Chiselling | | | | Inclination and Orientation | | | | Drilling Flush | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Depth Base | Diameter | Depth Base | Diameter | Depth Top | Depth Base | Duration | Tool | Depth Top | Depth Base | Inclination | Orientation | Depth Top | Depth Base | Type | Colour | Min (%) | Max (%) |
| 16.00<br>21.00 | 139<br>115 | 16.00 | 139 | | | | | | | | | 0.00 | 21.00 | Water | | | 100 |

Remarks

Borehole advanced to 21.0m using rotary coring drilling techniques to collect rock core samples and terminated on target depth. Borehole fitted with wellpoint upon completion. # Description based on drillers records. FI of 25 = Non-Intact.

# PHOENIX DRILLING LIMITED




| Contract No. | Contract | Picture Number |
|---|---|---|
| 4633 | Causeway Side | BH01 |

# PHOENIX DRILLING LIMITED




| Contract No. | Contract | Picture Number |
|---|---|---|
| 4633 | Causeway Side | BH01 |

# PHOENIX DRILLING LIMITED



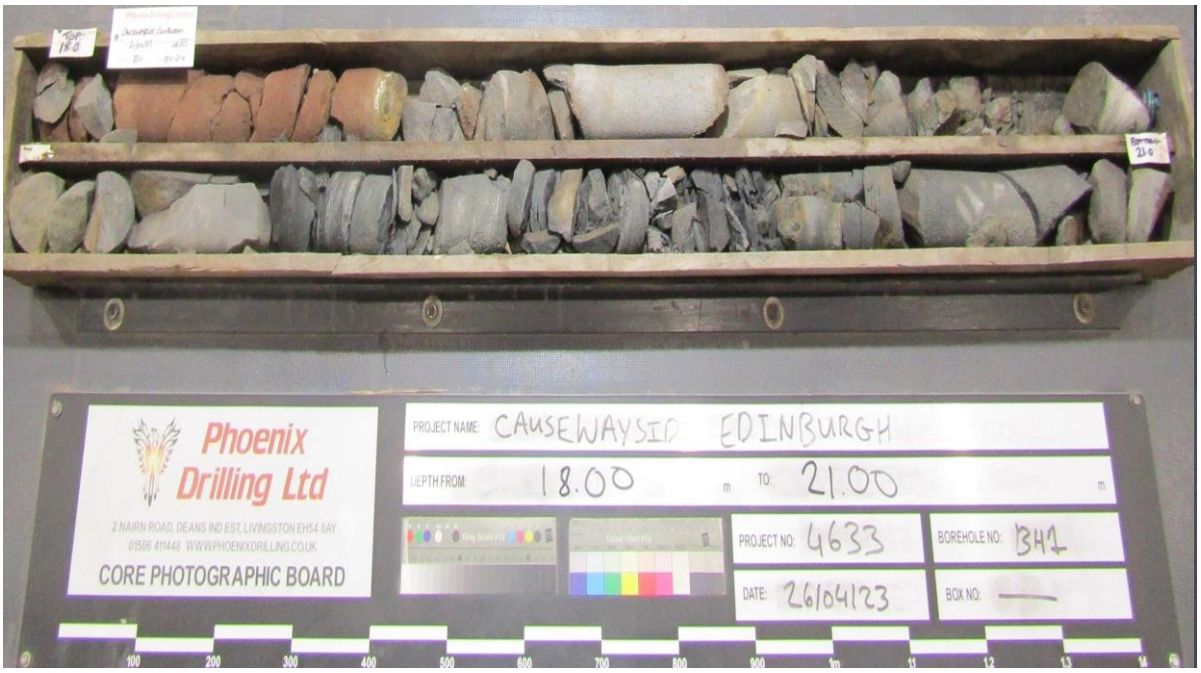

| Contract No. | Contract | Picture Number |
|---|---|---|
| 4633 | Causeway Side | BH01 |

# PHOENIX DRILLING LIMITED




| Contract No. | Contract | Picture Number |
|---|---|---|
| 4633 | Causeway Side | BH01 |

# PHOENIX DRILLING LIMITED




| Contract No. | Contract | Picture Number |
|---|---|---|
| 4633 | Causeway Side | BH01 |

# Rotary Core Log

Phoenix Drilling Ltd

| | | |
|---|---|---|
| Project Name: Causeway Side | Client: Mason Evans Partnership | Date: 26/04/2023 |
| Location: Edinburgh | Contractor: Phoenix Drilling Limited | |
| Project No. : 4633 | Crew Name: JH | Drilling Equipment: Fraste XL |

| Borehole Number | Hole Type | Level | Logged By | Scale | Page Number |
|---|---|---|---|---|---|
| RBH02A | RC | | CG | 1:50 | Sheet 1 of 3 |

| Well | Water | Depth (m) | Type /FI | Coring TCR | Coring SCR | Coring RQD | Diameter Recovery (SPT) | Depth (m) | Level (m) | Legend | Stratum Description |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | Carried out by cable percussion rig. |
| | | 4.00 | SPT | | | | (41) | 4.00 | | | # Very stiff locally stiff sandy gravelly CLAY with boulders. |
| | | 5.50 | SPT | | | | | | | | |
| | | 7.00 | SPT | | | | (23) | | | | |
| | | 8.50 | SPT | | | | | | | | |
| | | 10.00 | SPT | | | | (40) | | | | |
| | | | Type/FI | TCR | SCR | RQD | D/R/(SPT) | | | | |

1 2 3 4 5 6 7 8 9 10

| Hole Diameter | | Casing Diameter | | Chiselling | | | | Inclination and Orientation | | | | Drilling Flush | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Depth Base | Diameter | Depth Base | Diameter | Depth Top | Depth Base | Duration | Tool | Depth Top | Depth Base | Inclination | Orientation | Depth Top | Depth Base | Type | Colour | Min (%) | Max (%) |
| 15.00<br>24.00 | 139<br>115 | 15.00 | 139 | | | | | | | | | 0.00 | 24.00 | Water | | | 70 |

Remarks

Borehole advanced to 24.0m using rotary coring drilling techniques to collect rock core samples and terminated on target depth. Borehole fitted with wellpoint upon completion. # Description based on drillers records. FI of 25 = Non-Intact.

# Rotary Core Log

Phoenix Drilling Ltd

| | | |
|---|---|---|
| Project Name: Causeway Side | Client: Mason Evans Partnership | Date: 26/04/2023 |
| Location: Edinburgh | Contractor: Phoenix Drilling Limited | |
| Project No. : 4633 | Crew Name: JH | Drilling Equipment: Fraste XL |

| Borehole Number | Hole Type | Level | Logged By | Scale | Page Number |
|---|---|---|---|---|---|
| RBH02A | RC | | CG | 1:50 | Sheet 2 of 3 |

| Well | Water | Depth (m) | Type /FI | Coring TCR | Coring SCR | Coring RQD | Diameter Recovery (SPT) | Depth (m) | Level (m) | Legend | Stratum Description |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 13.00 | SPT | | | | (42) | | | | # Very stiff locally stiff sandy gravelly CLAY with boulders. |
| | | 15.00 - 16.50 | 25 | 100 | 56 | 40 | | 15.50 | | | |
| | | | 6 | | | | | 15.65 | | | Very weak reddish brown thinly laminated MUDSTONE. Heavily weathered and recovered predominantly non-intact. |
| | | | 10 | | | | | 16.25 | | | Moderately strong whitish grey thickly bedded crystalline fine grained LIMESTONE. Fractures are 0-45 degrees closely spaced undulating rough. Lightly to locally moderately weathered with a reddish sandy clay staining throughout. |
| | | 16.50 - 19.50 | 25 | 20 | 3 | 0 | | | | | Very weak reddish brown thinly laminated MUDSTONE. Heavily weathered and recovered predominantly non-intact. |
| | | | 0 | | | | | | | | *Assumed Zone of Core Loss (AZCL).* |
| | | | 25 | | | | | | | | |
| | | | | | | | | 19.50 | | | Weak reddish brown thinly laminated MUDSTONE. Fractures are 0-90 degrees |
| | | | Type/FI | TCR | SCR | RQD | D/R/(SPT) | | | | |

| Hole Diameter | | Casing Diameter | | Chiselling | | | | Inclination and Orientation | | | | Drilling Flush | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Depth Base | Diameter | Depth Base | Diameter | Depth Top | Depth Base | Duration | Tool | Depth Top | Depth Base | Inclination | Orientation | Depth Top | Depth Base | Type | Colour | Min (%) | Max (%) |
| 15.00<br>24.00 | 139<br>115 | 15.00 | 139 | | | | | | | | | 0.00 | 24.00 | Water | | | 70 |

Remarks

Borehole advanced to 24.0m using rotary coring drilling techniques to collect rock core samples and terminated on target depth. Borehole fitted with wellpoint upon completion. # Description based on drillers records. FI of 25 = Non-Intact.

# Rotary Core Log

Phoenix Drilling Ltd

| | | |
|---|---|---|
| Project Name: Causeway Side | Client: Mason Evans Partnership | Date: 26/04/2023 |
| Location: Edinburgh | Contractor: Phoenix Drilling Limited | |
| Project No. : 4633 | Crew Name: JH | Drilling Equipment: Fraste XL |

| Borehole Number | Hole Type | Level | Logged By | Scale | Page Number |
|---|---|---|---|---|---|
| RBH02A | RC | | CG | 1:50 | Sheet 3 of 3 |

| Well | Water | Depth (m) | Type /FI | Coring TCR | Coring SCR | Coring RQD | Diameter Recovery (SPT) | Depth (m) | Level (m) | Legend | Stratum Description |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 19.50 - 22.00 | 25 | 60 | 28 | 0 | | 21.30 | | | Weak reddish brown thinly laminated MUDSTONE. Fractures are 0-90 degrees closely spaced undulating rough. Lightly weathered with some locally heavily weathered sections recovered predominantly non-intact. |
| | | | 5 | | | | | 21.40 | | | |
| | | | 1 | | | | | 21.70 | | | Moderately strong whitish grey thickly bedded crystalline fine grained LIMESTONE. Fractures are 0-20 degrees closely spaced undulating rough. Lightly to locally moderately weathered with a reddish sandy clay staining throughout. |
| | | | 25 | | | | | 22.00 | | | |
| | | | 4 | | | | | 22.50 | | | Weak dark grey thinly laminated MUDSTONE. Very heavily weathered and recovered non-intact as an angular gravel. |
| | | 22.00 - 24.00 | 15 | 50 | 35 | 7 | | 22.70 | | | Moderately strong whitish grey thickly bedded crystalline fine grained LIMESTONE. Fractures are 0-20 degrees closely spaced undulating rough. Lightly weathered throughout. |
| | | | 1 | | | | | 24.00 | | | Weak dark grey thinly laminated micaceous MUDSTONE. Fractures are 0-40 degrees closely spaced undulating smooth. Lightly weathered throughout with some sandy infilling. Recovered badly fractures. |
| | | | 25 | | | | | | | | Moderately strong whitish grey thickly bedded crystalline fine grained LIMESTONE. Fractures are 0-20 degrees widely spaced undulating rough. Lightly weathered throughout. |
| | | | | | | | | | | | Weak dark grey thinly laminated MUDSTONE. Moderately weathered and recovered predominantly non-intact. |
| | | | | | | | | | | | *AZCL.* |
| | | | | | | | | | | | End of Borehole at 24.000m |
| | | | Type/FI | TCR | SCR | RQD | D/R/(SPT) | | | | |

| Hole Diameter | | Casing Diameter | | Chiselling | | | | Inclination and Orientation | | | | Drilling Flush | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Depth Base | Diameter | Depth Base | Diameter | Depth Top | Depth Base | Duration | Tool | Depth Top | Depth Base | Inclination | Orientation | Depth Top | Depth Base | Type | Colour | Min (%) | Max (%) |
| 15.00<br>24.00 | 139<br>115 | 15.00 | 139 | | | | | | | | | 0.00 | 24.00 | Water | | | 70 |

Remarks

Borehole advanced to 24.0m using rotary coring drilling techniques to collect rock core samples and terminated on target depth. Borehole fitted with wellpoint upon completion. # Description based on drillers records. FI of 25 = Non-Intact.

# PHOENIX DRILLING LIMITED




| Contract No. | Contract | Picture Number |
|---|---|---|
| 4633 | Causeway Side | BH02 |

# PHOENIX DRILLING LIMITED




| Contract No. | Contract | Picture Number |
|---|---|---|
| 4633 | Causeway Side | BH02 |

# PHOENIX DRILLING LIMITED




| Contract No. | Contract | Picture Number |
|---|---|---|
| 4633 | Causeway Side | BH02 |

# PHOENIX DRILLING LIMITED




| Contract No. | Contract | Picture Number |
|---|---|---|
| 4633 | Causeway Side | BH02 |

# Rotary Core Log

Phoenix Drilling Ltd

| | | |
|---|---|---|
| Project Name: Causeway Side | Client: Mason Evans Partnership | Date: 27/04/2023 |
| Location: Edinburgh | Contractor: Phoenix Drilling Limited | |
| Project No. : 4633 | Crew Name: JH | Drilling Equipment: Fraste XL |

| Borehole Number | Hole Type | Level | Logged By | Scale | Page Number |
|---|---|---|---|---|---|
| RBH03 | RC | | CG | 1:50 | Sheet 1 of 3 |

| Well | Water | Depth (m) | Type /FI | Coring TCR | Coring SCR | Coring RQD | Diameter Recovery (SPT) | Depth (m) | Level (m) | Legend | Stratum Description |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | Carried out by cable percussion rig. |
| | | | | | | | | 2.00 | | | Very stiff locally stiff sandy gravelly CLAY with cobbles and boulders. |
| | | 2.50 | SPT | | | | (34) | | | | |
| | | 4.00 | SPT | | | | (26) | | | | |
| | | 5.50 | SPT | | | | (39) | | | | |
| | | 7.00 | SPT | | | | (40) | | | | |
| | | 8.50 | SPT | | | | (37) | | | | |
| | | 10.00 | SPT | | | | (38) | | | | |
| | | | Type/FI | TCR | SCR | RQD | D/R/(SPT) | | | | |

| Hole Diameter | | Casing Diameter | | Chiselling | | | | Inclination and Orientation | | | | Drilling Flush | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Depth Base | Diameter | Depth Base | Diameter | Depth Top | Depth Base | Duration | Tool | Depth Top | Depth Base | Inclination | Orientation | Depth Top | Depth Base | Type | Colour | Min (%) | Max (%) |
| 16.00 | 139 | 16.00 | 139 | | | | | | | | | 0.00 | 14.50 | Air/Mist | | | 100 |
| 25.00 | 115 | | | | | | | | | | | 14.50 | 16.00 | Air | | | 100 |
| | | | | | | | | | | | | 16.00 | 25.00 | Water | | | 100 |

Remarks

Borehole advanced to 25.0m using rotary coring drilling techniques to collect rock core samples and terminated on target depth. Borehole fitted with wellpoint upon completion. # Description based on drillers records. FI of 25 = Non-Intact.

# Rotary Core Log

Phoenix Drilling Ltd

| | | |
|---|---|---|
| Project Name: Causeway Side | Client: Mason Evans Partnership | Date: 27/04/2023 |
| Location: Edinburgh | Contractor: Phoenix Drilling Limited | |
| Project No. : 4633 | Crew Name: JH | Drilling Equipment: Fraste XL |

| Borehole Number | Hole Type | Level | Logged By | Scale | Page Number |
|---|---|---|---|---|---|
| RBH03 | RC | | CG | 1:50 | Sheet 2 of 3 |

| Well | Water | Depth (m) | Type /FI | Coring TCR | Coring SCR | Coring RQD | Diameter Recovery (SPT) | Depth (m) | Level (m) | Legend | Stratum Description |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 11.50 | SPT | | | | | | | | Very stiff locally stiff sandy gravelly CLAY with cobbles and boulders. |
| | | 13.00 | SPT | | | | (44) | | | | |
| | ▼ | 16.00 - 19.00 | 11<br>4<br>7 | 50 | 50 | 12 | | 16.00<br>16.70<br>17.10 | | | Weak reddish brown thinly laminated MUDSTONE. Fractures are 0-25 degrees closely spaced undulating smooth. Lightly weathered throughout with a rare brownish grey clayey coating.<br>Moderately strong bluish grey thickly bedded fine grained crystalline LIMESTONE. Fractures are 0-10 degrees closely spaced undulating rough. Lightly weathered with a grey sandy infill.<br>Weak reddish brown thinly laminated MUDSTONE. Heavily weathered with large sections of no recovery, and non-intact.<br>*Assumed Zone of Core Loss (AZCL).* |
| | | | Type/FI | TCR | SCR | RQD | D/R/(SPT) | | | | |

11 12 13 14 15 16 17 18 19 20

| Hole Diameter | | Casing Diameter | | Chiselling | | | | Inclination and Orientation | | | | Drilling Flush | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Depth Base | Diameter | Depth Base | Diameter | Depth Top | Depth Base | Duration | Tool | Depth Top | Depth Base | Inclination | Orientation | Depth Top | Depth Base | Type | Colour | Min (%) | Max (%) |
| 16.00 | 139 | 16.00 | 139 | | | | | | | | | 0.00 | 14.50 | Air/Mist | | | 100 |
| 25.00 | 115 | | | | | | | | | | | 14.50 | 16.00 | Air | | | 100 |
| | | | | | | | | | | | | 16.00 | 25.00 | Water | | | 100 |

Remarks

Borehole advanced to 25.0m using rotary coring drilling techniques to collect rock core samples and terminated on target depth. Borehole fitted with wellpoint upon completion. # Description based on drillers records. FI of 25 = Non-Intact.

# Rotary Core Log

Phoenix Drilling Ltd

| | | |
|---|---|---|
| Project Name: Causeway Side | Client: Mason Evans Partnership | Date: 27/04/2023 |
| Location: Edinburgh | Contractor: Phoenix Drilling Limited | |
| Project No. : 4633 | Crew Name: JH | Drilling Equipment: Fraste XL |

| Borehole Number | Hole Type | Level | Logged By | Scale | Page Number |
|---|---|---|---|---|---|
| RBH03 | RC | | CG | 1:50 | Sheet 3 of 3 |

| Well | Water | Depth (m) | Type /FI | Coring TCR | Coring SCR | Coring RQD | Diameter Recovery (SPT) | Depth (m) | Level (m) | Legend | Stratum Description |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 19.00 - 22.00 | 8 | 63 | 50 | 20 | | 20.65 | | | Weak reddish brown thinly laminated MUDSTONE. Heavily weathered with large sections of no recovery, and non-intact. |
| | | | 10 | | | | | | | | Weak dark grey thinly laminated micaceous MUDSTONE. Fractures are 0-30 degrees closely spaced undulating smooth. Lightly weathered throughout with some sandy infilling. |
| | | | 25 | | | | | 21.60 | | | |
| | | | 7 | | | | | | | | Moderately strong bluish grey thickly bedded fine grained crystalline LIMESTONE. Fractures are 0-10 degrees closely spaced undulating rough. Lightly weathered with a grey sandy infill. |
| | | 22.00 - 25.00 | | 40 | 40 | 33 | | | | | *AZCL.* |
| | | | 5 | | | | | 23.70 | | | Hard white speckled orange fine to medium grained carbonaceous SANDSTONE. Fractures are 0-15 degrees closely spaced undulating rough. Fresh to locally lightly weathered. |
| | | | | | | | | 25.00 | | | End of Borehole at 25.000m |
| | | | Type/FI | TCR | SCR | RQD | D/R/(SPT) | | | | |

21 22 23 24 25 26 27 28 29 30

| Hole Diameter | | Casing Diameter | | Chiselling | | | | Inclination and Orientation | | | | Drilling Flush | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Depth Base | Diameter | Depth Base | Diameter | Depth Top | Depth Base | Duration | Tool | Depth Top | Depth Base | Inclination | Orientation | Depth Top | Depth Base | Type | Colour | Min (%) | Max (%) |
| 16.00 | 139 | 16.00 | 139 | | | | | | | | | 0.00 | 14.50 | Air/Mist | | | 100 |
| 25.00 | 115 | | | | | | | | | | | 14.50 | 16.00 | Air | | | 100 |
| | | | | | | | | | | | | 16.00 | 25.00 | Water | | | 100 |

Remarks

Borehole advanced to 25.0m using rotary coring drilling techniques to collect rock core samples and terminated on target depth. Borehole fitted with wellpoint upon completion. # Description based on drillers records. FI of 25 = Non-Intact.

# PHOENIX DRILLING LIMITED




| Contract No. | Contract | Picture Number |
|---|---|---|
| 4633 | Causeway Side | BH03 |

# PHOENIX DRILLING LIMITED




| Contract No. | Contract | Picture Number |
|---|---|---|
| 4633 | Causeway Side | BH03 |

# PHOENIX DRILLING LIMITED




| Contract No. | Contract | Picture Number |
|---|---|---|
| 4633 | Causeway Side | BH03 |

**Appendix 7**

**Gas and Groundwater Monitoring Results**

MASON EVANS
Geo-Environmental Consultants

# Gas and Groundwater Monitoring Results

| | |
|---|---|
| **Project Number:** | P23-075 |
| **Site:** | 140 Causewayside, Edinburgh |
| **Date:** | 10.05.23 |
| **Readings taken by:** | FC |

| **Background Data** | |
|---|---|
| **Weather Conditions** | Rain |
| **Ground Conditions (dry/wet):** | Wet |
| **Air Temperature (°C)** | 14 |
| **Atmospheric Pressure (mB) (start):** | 998 |
| **Atmospheric Pressure (mB) (finish):** | 997 |
| **O2 (%)** | 20.5 |
| **Time (hh:mm)** | 13:30 |

| **Borehole No.** | **Interval (seconds)** | **Gas** | | | | | | | | | | | | **Groundwater** | | **Sampling** | | | **PID Results** |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | **Oxygen $O_2$ (%)** | | **Carbon Dioxide $CO_2$ (%)** | | **Methane $CH_4$ (%)** | | **Hydrogen Sulphide $H_2S$ (PPM)** | | **Carbon Monoxide CO (PPM)** | | **Lower Explosive Limit (LEL)** | **Flow (l/hr)** | | **Wd[3] (mbgl)** | **DoW[4] (mbgl)** | **Pr[5]** | **R[6]** | **S[7]** | |
| | | **P[1]** | **SS[2]** | **P[1]** | **SS[2]** | **P[1]** | **SS[2]** | **P[1]** | **SS[2]** | **P[1]** | **SS[2]** | | **P[1]** | **SS[2]** | | | | | | |
| BH01B/RBH01B | 30 | | 10.8 | | 0.6 | | 0.0 | | 0.0 | | - | 0.0 | 10.2 | 0.0 | 1.00 | 14.40 | 3 | Y | Y | |
| | 60 | | 12.4 | | 0.6 | | 0.0 | | 0.0 | | - | 0.0 | | | | | | | | |
| | 90 | | 12.9 | | 0.6 | | 0.0 | | 0.0 | | - | 0.0 | | | | | | | | |
| | 120 | | 13.2 | | 0.6 | | 0.0 | | 0.0 | | - | 0.0 | | | | | | | | |
| BH02A/RBH02A | 30 | | - | | - | | - | | - | | - | - | | | 1.00 | 9.30 | 3 | Y | Y | |
| | 60 | | - | | - | | - | | - | | - | - | | | | | | | | |
| | 90 | | - | | - | | - | | - | | - | - | | | | | | | | |
| | 120 | | - | | - | | - | | - | | - | - | | | | | | | | |
| BH03/RBH03 | 30 | | 8.3 | | 0.2 | | 0.0 | | 0.0 | | - | 0.0 | 56.4 | 0.0 | 1.35 | 11.60 | | | | |
| | 60 | | 8.4 | | 0.2 | | 0.0 | | 0.0 | | - | 0.0 | | | | | | | | |
| | 90 | | 8.4 | | 0.2 | | 0.0 | | 0.0 | | - | 0.0 | | | | | | | | |
| | 120 | | 8.4 | | 0.2 | | 0.0 | | 0.0 | | - | 0.0 | | | | | | | | |

| | |
|---|---|
| **Remarks** | BH02A/RBH02A: Gas tap left open so couldn't monitore gas concentrations BH03/RBH03 - Bailer caught in standpipe after 26 out of 62 purges required |
| **Borehole Damage Record/ Installation Record** | |
| **Borehole Condition Statement** | |

Key:
1 – Peak
2 – Steady state
3 – Groundwater depth
4 – Depth of well
5 –Purged well volumes
6 – Recharge (yes/no)
7 – Sampled (yes/no)

| **Gas Monitor Model:** | **Serial No:** | **Recalibration Due:** |
|---|---|---|
| GFM430 | 10309 | 11.06.24 |

MASON EVANS
Geo-Environmental Consultants

# Gas and Groundwater Monitoring Results

| | | |
|---|---|---|
| **Project Number:** | P23-075 | |
| **Site:** | Causewayside | |
| **Date:** | 24.05.23 | |
| **Readings taken by:** | FC | |
| **Background Data** | **Weather Conditions** | Sunny |
| | **Ground Conditions (dry/wet):** | Dry |
| | **Air Temperature (°C)** | 15 |
| | **Atmospheric Pressure (mB) (start):** | 1016 |
| | **Atmospheric Pressure (mB) (finish):** | 1015 |
| | **O2 (%)** | 20.5 |
| | **Time (hh:mm)** | 10:20 |

| **Borehole No.** | **Interval (seconds)** | **Gas** | | | | | | | | | | | | | **Groundwater** | | **Sampling** | | | **PID Results** |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | **Oxygen $O_2$ (%)** | | **Carbon Dioxide $CO_2$ (%)** | | **Methane $CH_4$ (%)** | | **Hydrogen Sulphide $H_2S$ (PPM)** | | **Carbon Monoxide CO (PPM)** | | **Lower Explosive Limit (LEL)** | **Flow (l/hr)** | | **Wd[3] (mbgl)** | **DoW[4] (mbgl)** | **Pr[5]** | **R[6]** | **S[7]** | |
| | | **P[1]** | **SS[2]** | **P[1]** | **SS[2]** | **P[1]** | **SS[2]** | **P[1]** | **SS[2]** | **P[1]** | **SS[2]** | | **P[1]** | **SS[2]** | | | | | | |
| BH01B/RBH01B | 30 | | 18.0 | | 0.4 | | 0.0 | | 0.0 | | - | 0.0 | 0.0 | 0.0 | 1.10 | 14.60 | 3 | Y | Y | |
| | 60 | | 18.0 | | 0.4 | | 0.0 | | 0.0 | | - | 0.0 | | | | | | | | |
| | 90 | | 18.0 | | 0.4 | | 0.0 | | 0.0 | | - | 0.0 | | | | | | | | |
| | 120 | | 18.0 | | 0.4 | | 0.0 | | 0.0 | | - | 0.0 | | | | | | | | |
| BH02A/RBH02A | 30 | | 17.2 | | 0.2 | | 0.0 | | 0.0 | | - | 0.0 | 3.2 | 0.0 | 1.55 | 9.00 | 3 | Y | Y | |
| | 60 | | 17.0 | | 0.2 | | 0.0 | | 0.0 | | - | 0.0 | | | | | | | | |
| | 90 | | 16.9 | | 0.2 | | 0.0 | | 0.0 | | - | 0.0 | | | | | | | | |
| | 120 | | 16.9 | | 0.2 | | 0.0 | | 0.0 | | - | 0.0 | | | | | | | | |
| BH03/RBH03 | 30 | | 9.0 | | 0.3 | | 0.0 | | 0.0 | | - | 0.0 | 2.6 | 0.0 | N/A | N/A | N | N | N | |
| | 60 | | 9.8 | | 0.3 | | 0.0 | | 0.0 | | - | 0.0 | | | | | | | | |
| | 90 | | 11.0 | | 0.2 | | 0.0 | | 0.0 | | - | 0.0 | | | | | | | | |
| | 120 | | 12.0 | | 0.2 | | 0.0 | | 0.0 | | - | 0.0 | | | | | | | | |
| | 150 | | 14.0 | | 0.2 | | 0.0 | | 0.0 | | - | 0.0 | | | | | | | | |

| | |
|---|---|
| **Remarks** | Bailer stuck in BH03/RBH03 so couldn't monitor groundwater levels or the depth to the base of the well |
| **Borehole Damage Record/ Installation Record** | |
| **Borehole Condition Statement** | |

Key:
1 – Peak
2 – Steady state
3 – Groundwater depth
4 – Depth of well
5 –Purged well volumes
6 – Recharge (yes/no)
7 – Sampled (yes/no)

| **Gas Monitor Model:** | **Serial No:** | **Recalibration Due:** |
|---|---|---|
| GFM430 | 10309 | 14.06.23 |

MASON EVANS
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# Gas and Groundwater Monitoring Results

| | |
|---|---|
| **Project Number:** | P23-075 |
| **Site:** | Causewayside |
| **Date:** | 12/06/2023 |
| **Readings taken by:** | DM |

| **Background Data** | |
|---|---|
| **Weather Conditions** | Cloudy |
| **Ground Conditions (dry/wet):** | Dry |
| **Air Temperature (°C)** | 19 |
| **Atmospheric Pressure (mB) (start):** | 1009 |
| **Atmospheric Pressure (mB) (finish):** | 1008 |
| **O2 (%)** | 20.9 |
| **Time (hh:mm)** | 11:15 |

| **Borehole No.** | **Interval (seconds)** | **Gas** | | | | | | | | | | | | | **Groundwater** | | **Sampling** | | | **PID Results** |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | **Oxygen $O_2$ (%)** | | **Carbon Dioxide $CO_2$ (%)** | | **Methane $CH_4$ (%)** | | **Hydrogen Sulphide $H_2S$ (PPM)** | | **Carbon Monoxide CO (PPM)** | | **Lower Explosive Limit (LEL)** | **Flow (l/hr)** | | **Wd[3] (mbgl)** | **DoW[4] (mbgl)** | **Pr[5]** | **R[6]** | **S[7]** | |
| | | **P[1]** | **SS[2]** | **P[1]** | **SS[2]** | **P[1]** | **SS[2]** | **P[1]** | **SS[2]** | **P[1]** | **SS[2]** | | **P[1]** | **SS[2]** | | | | | | |
| BH01B/RBH01B | 30 | - | 17.9 | - | 0.6 | - | 0.0 | - | 0.0 | - | 0.0 | 0.0 | 3.2 | 0.0 | 1.25 | 13.90 | N/A | N/A | Y | N/A |
| | 60 | - | 17.8 | - | 0.6 | - | 0.0 | - | 0.0 | - | 0.0 | 0.0 | | | | | | | | |
| | 90 | - | 17.8 | - | 0.6 | - | 0.0 | - | 0.0 | - | 0.0 | 0.0 | | | | | | | | |
| | 120 | - | 17.8 | - | 0.6 | - | 0.0 | - | 0.0 | - | 0.0 | 0.0 | | | | | | | | |
| BH02A/RBH02A | 30 | - | 16.5 | - | 0.4 | - | 0.0 | - | 0.0 | - | 0.0 | 0.0 | 22.1 | 0.0 | 1.50 | 8.80 | 3 | Y | Y | N/A |
| | 60 | - | 16.5 | - | 0.4 | - | 0.0 | - | 0.0 | - | 0.0 | 0.0 | | | | | | | | |
| | 90 | - | 16.5 | - | 0.4 | - | 0.0 | - | 0.0 | - | 0.0 | 0.0 | | | | | | | | |
| | 120 | - | 16.5 | - | 0.4 | - | 0.0 | - | 0.0 | - | 0.0 | 0.0 | | | | | | | | |
| BH03/R03 | 30 | - | 15.5 | - | 0.2 | - | 0.0 | - | 0.0 | - | 0.0 | 0.0 | 0.0 | 0.0 | N/A | N/A | N | N | N | N/A |
| | 60 | - | 15.7 | - | 0.2 | - | 0.0 | - | 0.0 | - | 0.0 | 0.0 | | | | | | | | |
| | 90 | - | 15.7 | - | 0.2 | - | 0.0 | - | 0.0 | - | 0.0 | 0.0 | | | | | | | | |
| | 120 | - | 15.7 | - | 0.2 | - | 0.0 | - | 0.0 | - | 0.0 | 0.0 | | | | | | | | |

| | |
|---|---|
| **Remarks** | BH01B/RBH0B - Strong sulphorous odour so sampled straight away<br>Bailer stuck in BH03/RBH03 so couldn't monitor groundwater levels or the depth to the base of the well |
| **Borehole Damage Record/ Installation Record** | |
| **Borehole Condition Statement** | |

Key:
1 – Peak
2 – Steady state
3 – Groundwater depth
4 – Depth of well
5 –Purged well volumes
6 – Recharge (yes/no)
7 – Sampled (yes/no)

| **Gas Monitor Model:** | **Serial No:** | **Recalibration Due:** |
|---|---|---|
| GFM436 | 12984 | 28.02.2024 |

MASON EVANS
Geo-Environmental Consultants

# Gas and Groundwater Monitoring Results

| | |
|---|---|
| **Project Number:** | P23-075 |
| **Site:** | Causewayside |
| **Date:** | 21/06/2023 |
| **Readings taken by:** | DM |

| **Background Data** | | |
|---|---|---|
| | **Weather Conditions** | Sunny |
| | **Ground Conditions (dry/wet):** | Dry |
| | **Air Temperature (°C)** | 18 |
| | **Atmospheric Pressure (mB) (start):** | 1000 |
| | **Atmospheric Pressure (mB) (finish):** | 999 |
| | **O2 (%)** | 20.6 |
| | **Time (hh:mm)** | 10:30 |

| **Borehole No.** | **Interval (seconds)** | **Gas** | | | | | | | | | | | | | **Groundwater** | | **Sampling** | | | **PID Results** |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | **Oxygen $O_2$ (%)** | | **Carbon Dioxide $CO_2$ (%)** | | **Methane $CH_4$ (%)** | | **Hydrogen Sulphide $H_2S$ (PPM)** | | **Carbon Monoxide CO (PPM)** | | **Lower Explosive Limit (LEL)** | **Flow (l/hr)** | | **Wd[3] (mbgl)** | **DoW[4] (mbgl)** | **Pr[5]** | **R[6]** | **S[7]** | |
| | | **P[1]** | **SS[2]** | **P[1]** | **SS[2]** | **P[1]** | **SS[2]** | **P[1]** | **SS[2]** | **P[1]** | **SS[2]** | | **P[1]** | **SS[2]** | | | | | | |
| BH01B/RBH01B | 30 | - | 14.8 | - | 1.8 | - | 0.0 | - | 0.0 | - | 0.0 | 0.0 | 18.7 | 0.0 | 1.55 | 14.00 | N | N | N | N/A |
| | 60 | - | 14.7 | - | 1.7 | - | 0.0 | - | 0.0 | - | 0.0 | 0.0 | | | | | | | | |
| | 90 | - | 14.7 | - | 1.7 | - | 0.0 | - | 0.0 | - | 0.0 | 0.0 | | | | | | | | |
| | 120 | - | 14.7 | - | 1.7 | - | 0.0 | - | 0.0 | - | 0.0 | 0.0 | | | | | | | | |
| BH02A/RBH02A | 30 | - | 17.7 | - | 0.4 | - | 0.0 | - | 0.0 | - | 0.0 | 0.0 | 13.6 | 0.0 | 1.50 | 8.80 | N | N | N | N/A |
| | 60 | - | 17.7 | - | 0.4 | - | 0.0 | - | 0.0 | - | 0.0 | 0.0 | | | | | | | | |
| | 90 | - | 17.7 | - | 0.4 | - | 0.0 | - | 0.0 | - | 0.0 | 0.0 | | | | | | | | |
| | 120 | - | 17.7 | - | 0.4 | - | 0.0 | - | 0.0 | - | 0.0 | 0.0 | | | | | | | | |
| BH03/RBH03 | 30 | - | 17.4 | - | 0.2 | - | 0.0 | - | 0.0 | - | 0.0 | 0.0 | 0.0 | 0.0 | N/A | N/A | N | N | N | N/A |
| | 60 | - | 17.7 | - | 0.2 | - | 0.0 | - | 0.0 | - | 0.0 | 0.0 | | | | | | | | |
| | 90 | - | 17.9 | - | 0.2 | - | 0.0 | - | 0.0 | - | 0.0 | 0.0 | | | | | | | | |
| | 120 | - | 17.9 | - | 0.2 | - | 0.0 | - | 0.0 | - | 0.0 | 0.0 | | | | | | | | |

| | |
|---|---|
| **Remarks** | Bailer stuck in BH03/RBH03 so couldn't monitor groundwater levels or the depth to the base of the well |
| **Borehole Damage Record/ Installation Record** | |
| **Borehole Condition Statement** | |

Key:
1 – Peak
2 – Steady state
3 – Groundwater depth
4 – Depth of well
5 –Purged well volumes
6 – Recharge (yes/no)
7 – Sampled (yes/no)

| **Gas Monitor Model:** | **Serial No:** | **Recalibration Due:** |
|---|---|---|
| GFM436 | 12984 | 28.02.2024 |

MASON EVANS
Geo-Environmental Consultants

# Gas and Groundwater Monitoring Results

| | |
|---|---|
| **Project Number:** | P23-075 |
| **Site:** | Causewayside |
| **Date:** | 06.07.2023 |
| **Readings taken by:** | CM |

| **Background Data** | |
|---|---|
| **Weather Conditions** | Clear |
| **Ground Conditions (dry/wet):** | Dry |
| **Air Temperature (°C)** | 13 |
| **Atmospheric Pressure (mB) (start):** | 997 |
| **Atmospheric Pressure (mB) (finish):** | 997 |
| **O2 (%)** | 20.7 |
| **Time (hh:mm)** | 1200-1330 |

| **Borehole No.** | **Interval (seconds)** | **Gas** | | | | | | | | | | | | | **Groundwater** | | **Sampling** | | | **PID Results** |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | **Oxygen $O_2$ (%)** | | **Carbon Dioxide $CO_2$ (%)** | | **Methane $CH_4$ (%)** | | **Hydrogen Sulphide $H_2S$ (PPM)** | | **Carbon Monoxide CO (PPM)** | | **Lower Explosive Limit (LEL)** | **Flow (l/hr)** | | **Wd[3] (mbgl)** | **DoW[4] (mbgl)** | **Pr[5]** | **R[6]** | **S[7]** | |
| | | **P[1]** | **SS[2]** | **P[1]** | **SS[2]** | **P[1]** | **SS[2]** | **P[1]** | **SS[2]** | **P[1]** | **SS[2]** | | **P[1]** | **SS[2]** | | | | | | |
| BH01B/RBH01B | 30 | - | 16.7 | - | 1.1 | - | 0.0 | - | 0.0 | - | 0.0 | 0.0 | 51.6 | 2.5 | 1.30 | 13.90 | 3 | Y | N | |
| | 60 | - | 16.7 | - | 1.1 | - | 0.0 | - | 0.0 | - | 0.0 | 0.0 | 2.5 | 2.5 | | | | | | |
| | 90 | - | 16.7 | - | 1.1 | - | 0.0 | - | 0.0 | - | 0.0 | 0.0 | 2.5 | 2.5 | | | | | | |
| | 120 | - | 16.7 | - | 1.1 | - | 0.0 | - | 0.0 | - | 0.0 | 0.0 | 2.5 | 2.5 | | | | | | |
| BH02A/RBH02A | 30 | - | 17.8 | - | 0.5 | - | 0.0 | - | 0.0 | - | 0.0 | 0.0 | 54.4 | 0.0 | 1.25 | 8.95 | 3 | Y | N | |
| | 60 | - | 17.8 | - | 0.5 | - | 0.0 | - | 0.0 | - | 0.0 | 0.0 | 0.0 | 0.0 | | | | | | |
| | 90 | - | 17.8 | - | 0.5 | - | 0.0 | - | 0.0 | - | 0.0 | 0.0 | 0.0 | 0.0 | | | | | | |
| | 120 | - | 17.8 | - | 0.5 | - | 0.0 | - | 0.0 | - | 0.0 | 0.0 | 0.0 | 0.0 | | | | | | |
| BH03/RBH03 | 30 | - | 16.8 | - | 0.4 | - | 0.0 | - | 0.0 | - | 0.0 | 0.0 | 0.0 | 0.0 | 1.00 | N/A | - | - | - | |
| | 60 | - | 16.8 | - | 0.4 | - | 0.0 | - | 0.0 | - | 0.0 | 0.0 | 0.0 | 0.0 | | | | | | |
| | 90 | - | 16.7 | - | 0.4 | - | 0.0 | - | 0.0 | - | 0.0 | 0.0 | 0.0 | 0.0 | | | | | | |
| | 120 | - | 16.7 | - | 0.4 | - | 0.0 | - | 0.0 | - | 0.0 | 0.0 | 0.0 | 0.0 | | | | | | |

| | |
|---|---|
| **Remarks** | Bailer stuck in BH03/RBH03 so couldn't monitor groundwater levels or the depth to the base of the well |
| **Borehole Damage Record/ Installation Record** | |
| **Borehole Condition Statement** | |

Key:
1 – Peak
2 – Steady state
3 – Groundwater depth
4 – Depth of well
5 –Purged well volumes
6 – Recharge (yes/no)
7 – Sampled (yes/no)

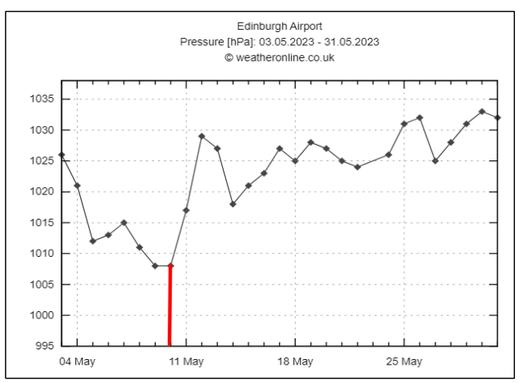
| **Gas Monitor Model:** | **Serial No:** | **Recalibration Due:** |
|---|---|---|
| GFM 436 | 12984 | 28.02.2024 |

# Ground Gas Assessment - Gas Screening Values (Carbon Dioxide)

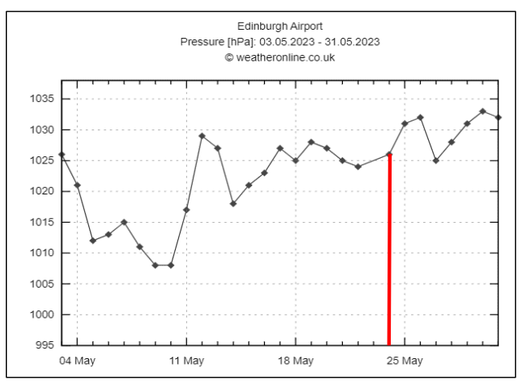
## Causewayside, Edinburgh

**P23-075**

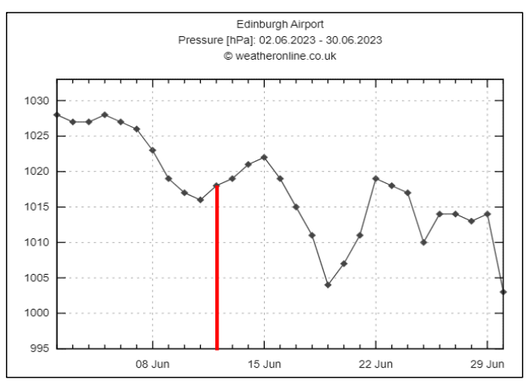
| Round 1 | | | | |
|---|---|---|---|---|
| 10.05.23 | | | | |
| Borehole No. | CO2 (%) | Recorded Flow (l/hr) | Flow Detection Value (l/hr) * | GSV (l/hr) (CO2) |
| BH01B/RBH01B | 0.6 | 0.0 | 0.1 | 0.0006 |
| BH02A/RBH02A | ** | ** | ** | ** |
| BH03/RBH03 | 0.2 | 0.0 | 0.1 | 0.0002 |



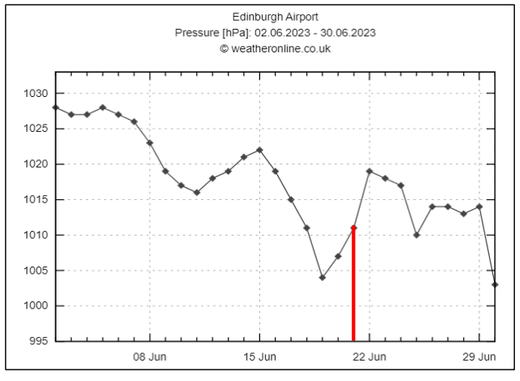

| Round 2 | | | | |
|---|---|---|---|---|
| 24.05.23 | | | | |
| Borehole No. | CO2 (%) | Recorded Flow (l/hr) | Flow Detection Value (l/hr) * | GSV (l/hr) (CO2) |
| BH01B/RBH01B | 0.4 | 0.0 | 0.1 | 0.0004 |
| BH02A/RBH02A | 0.2 | 0.0 | 0.1 | 0.0002 |
| BH03/RBH03 | 0.2 | 0.0 | 0.1 | 0.0002 |



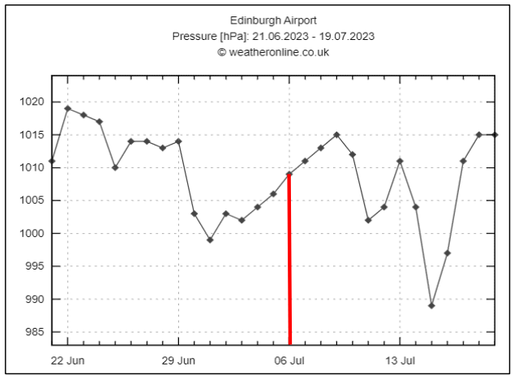

| Round 3 | | | | |
|---|---|---|---|---|
| 12.06.23 | | | | |
| Borehole No. | CO2 (%) | Recorded Flow (l/hr) | Flow Detection Value (l/hr) * | GSV (l/hr) (CO2) |
| BH01B/RBH01B | 0.6 | 0.0 | 0.1 | 0.0006 |
| BH02A/RBH02A | 0.4 | 0.0 | 0.1 | 0.0004 |
| BH03/RBH03 | 0.2 | 0.0 | 0.1 | 0.0002 |




| Round 4 | | | | |
|---|---|---|---|---|
| 21.06.23 | | | | |
| Borehole No. | CO2 (%) | Recorded Flow (l/hr) | Flow Detection Value (l/hr) * | GSV (l/hr) (CO2) |
| BH01B/RBH01B | 1.7 | 0.0 | 0.1 | 0.0017 |
| BH02A/RBH02A | 0.4 | 0.0 | 0.1 | 0.0004 |
| BH03/RBH03 | 0.2 | 0.0 | 0.1 | 0.0002 |




| Round 5 | | | | |
|---|---|---|---|---|
| 06.07.23 | | | | |
| Borehole No. | CO2 (%) | Recorded Flow (l/hr) | Flow Detection Value (l/hr) * | GSV (l/hr) (CO2) |
| BH01B/RBH01B | 1.1 | 2.5 | 2.5 | 0.0275 |
| BH02A/RBH02A | 0.5 | 0.0 | 0.1 | 0.0005 |
| BH03/RBH03 | 0.4 | 0.0 | 0.1 | 0.0004 |




***Note that the results were assessed once steady state was reached***

*** Flow detection value 0.1 l/hr if no flow detected or flow value recorded**

**** Borehole not monitorable (gas tap left open)**

**Calculated GSV corresponds to 'Characteristic Situation 1' (i.e. <0.07 l/hr)**

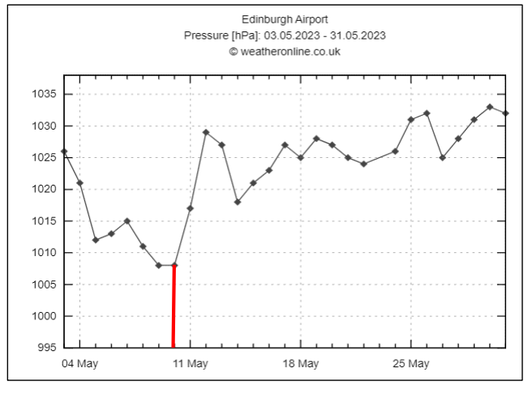
**Calculated GSV corresponds to 'Characteristic Situation 2' (i.e. 0.07 l/hr to <0.7 l/hr)**

# Ground Gas Assessment - Gas Screening Values (Methane)

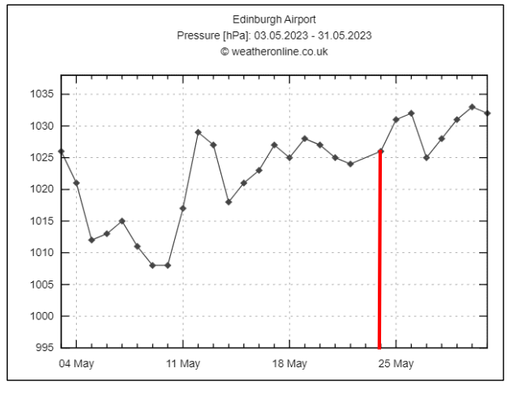
## Causewayside, Edinburgh

**P23-075**

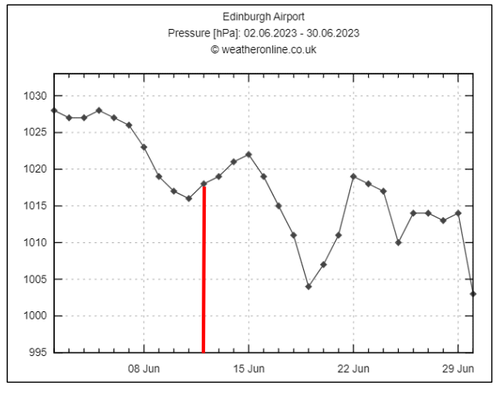
| Round 1 | | | | |
|---|---|---|---|---|
| 10.05.23 | | | | |
| **Borehole No.** | **CH4 (%)** | **Recorded Flow (l/hr)** | **Flow Detection Value (l/hr) *** | **GSV (l/hr) (CH4)** |
| BH01B/RBH01B | 0.0 | 0.0 | 0.1 | 0.0000 |
| BH02A/RBH02A | ** | ** | ** | ** |
| BH03/RBH03 | 0.0 | 0.0 | 0.1 | 0.0000 |



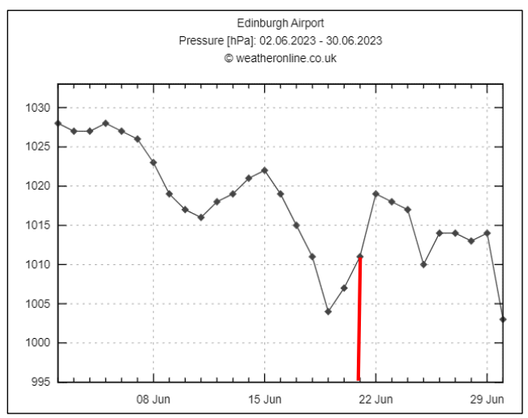

| Round 2 | | | | |
|---|---|---|---|---|
| 24.05.23 | | | | |
| **Borehole No.** | **CH4 (%)** | **Recorded Flow (l/hr)** | **Flow Detection Value (l/hr) *** | **GSV (l/hr) (CH4)** |
| BH01B/RBH01B | 0.0 | 0.0 | 0.1 | 0.0000 |
| BH02A/RBH02A | 0.0 | 0.0 | 0.1 | 0.0000 |
| BH03/RBH03 | 0.0 | 0.0 | 0.1 | 0.0000 |



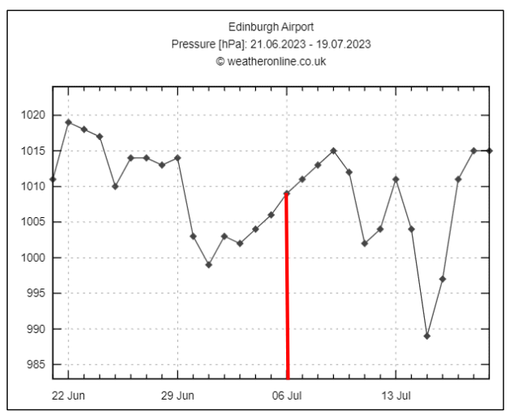

| Round 3 | | | | |
|---|---|---|---|---|
| 12.06.23 | | | | |
| **Borehole No.** | **CH4 (%)** | **Recorded Flow (l/hr)** | **Flow Detection Value (l/hr) *** | **GSV (l/hr) (CH4)** |
| BH01B/RBH01B | 0.0 | 0.0 | 0.1 | 0.0000 |
| BH02A/RBH02A | 0.0 | 0.0 | 0.1 | 0.0000 |
| BH03/RBH03 | 0.0 | 0.0 | 0.1 | 0.0000 |




| Round 4 | | | | |
|---|---|---|---|---|
| 21.06.23 | | | | |
| **Borehole No.** | **CH4 (%)** | **Recorded Flow (l/hr)** | **Flow Detection Value (l/hr) *** | **GSV (l/hr) (CH4)** |
| BH01B/RBH01B | 0.0 | 0.0 | 0.1 | 0.0000 |
| BH02A/RBH02A | 0.0 | 0.0 | 0.1 | 0.0000 |
| BH03/RBH03 | 0.0 | 0.0 | 0.1 | 0.0000 |




| Round 5 | | | | |
|---|---|---|---|---|
| 06.07.23 | | | | |
| **Borehole No.** | **CH4 (%)** | **Recorded Flow (l/hr)** | **Flow Detection Value (l/hr) *** | **GSV (l/hr) (CH4)** |
| BH01B/RBH01B | 0.0 | 2.5 | 2.5 | 0.0000 |
| BH02A/RBH02A | 0.0 | 0.0 | 0.1 | 0.0000 |
| BH03/RBH03 | 0.0 | 0.0 | 0.1 | 0.0000 |




***Note that the results were assessed once steady state was reached***

*** Flow detection value 0.1 l/hr if no flow detected or flow value recorded**

**** Borehole could not be located/not monitorable**

**Calculated GSV corresponds to 'Characteristic Situation 1' (i.e. <0.07 l/hr)**

**Calculated GSV corresponds to 'Characteristic Situation 2' (i.e. 0.07 l/hr to <0.7 l/hr)**